

Spring 2011

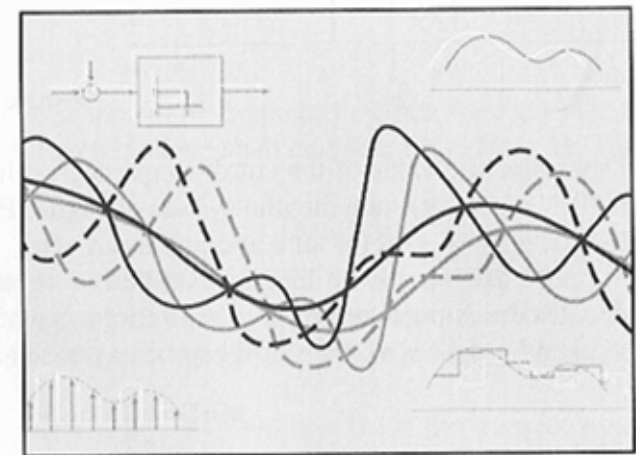
信號與系統 Signals and Systems

Chapter SS-7 Sampling

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NTU-EE

Feb11 – Jun11



Figures and images used in these lecture notes are adopted from
“Signals & Systems” by Alan V. Oppenheim and Alan S. Willsky, 1997

Introduction

(Chap 1)

LTI & Convolution

(Chap 2)

Bounded/Convergent

Periodic

FS

(Chap 3)

– CT
– DT

Aperiodic

FT

– CT (Chap 4)
– DT (Chap 5)

Unbounded/Non-convergent

LT

– CT (Chap 9)

zT

– DT (Chap 10)

Time-Frequency (Chap 6)

CT-DT (Chap 7)

Communication (Chap 8)

Control (Chap 11)

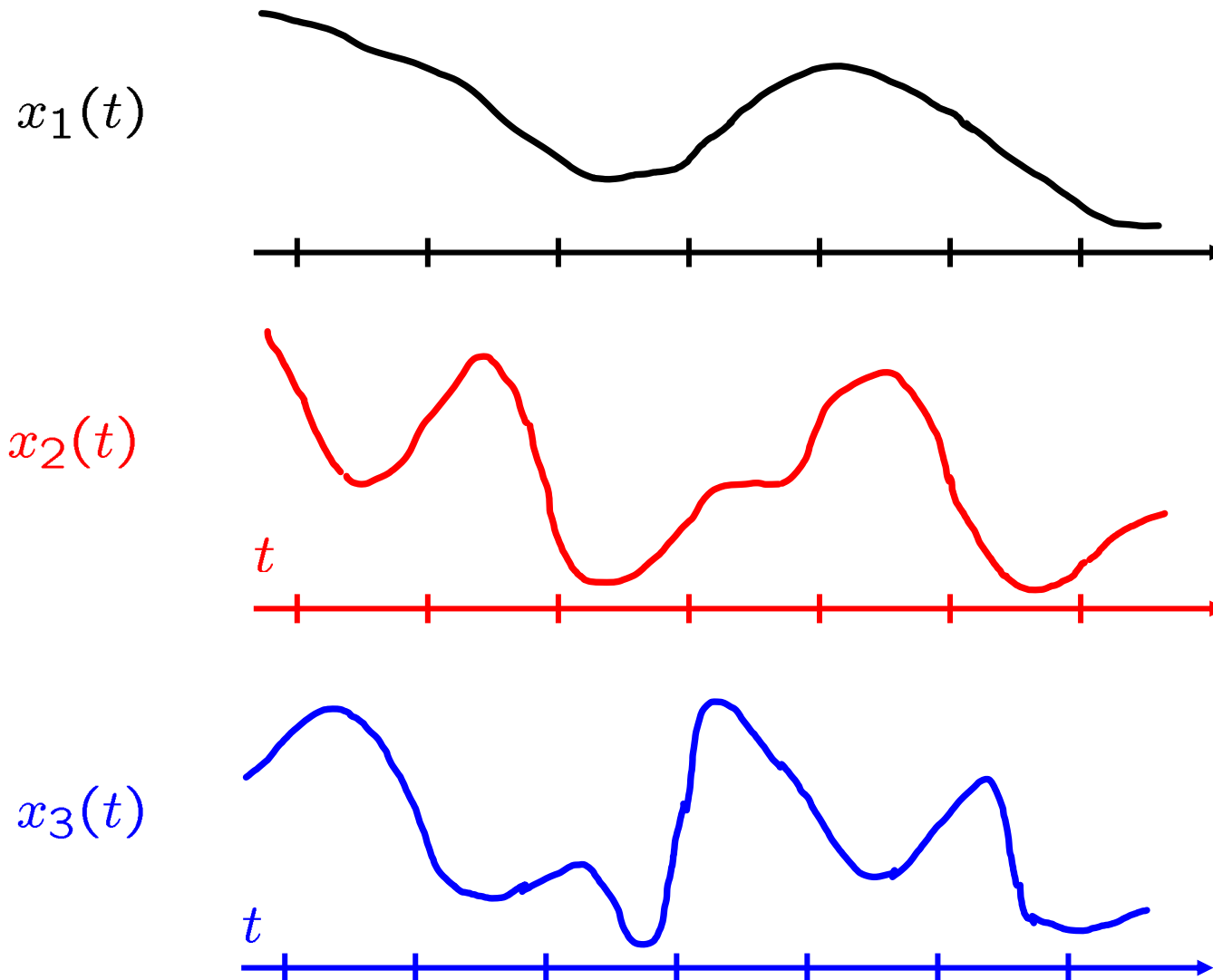
Digital
Signal
Processing (dsp-8)

Fourier Series, Fourier Transform, Laplace Transform, z-Transform

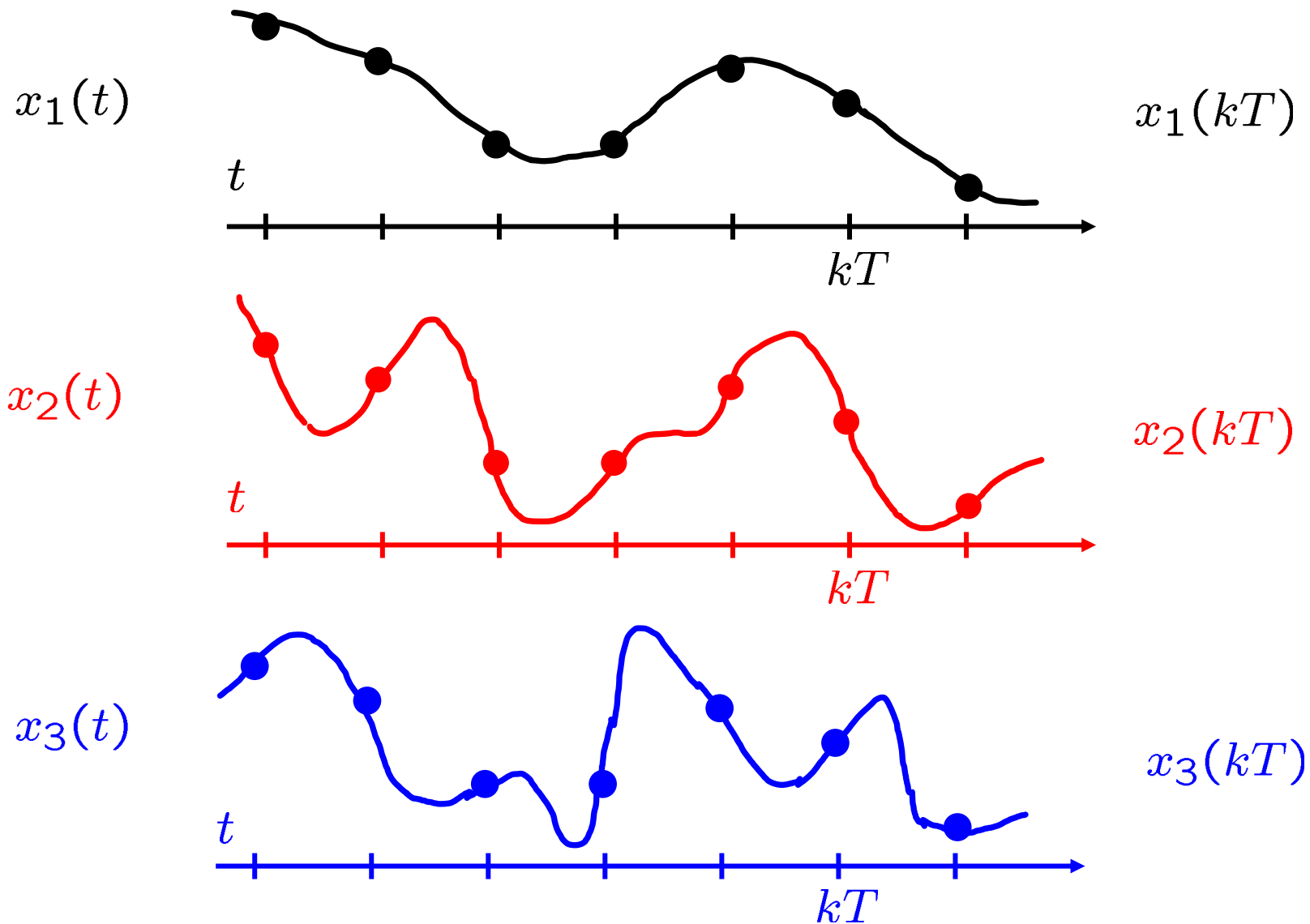
	CT		DT	
	time	frequency	time	frequency
FS				
FT				
LT/zT				

- Representation of of a Continuous-Time Signal by Its Samples: The Sampling Theorem
- Reconstruction of of a Signal from Its Samples Using Interpolation
- The Effect of Under-sampling: Aliasing
- Discrete-Time Processing of Continuous-Time Signals
- Sampling of Discrete-Time Signals

■ Representation of CT Signals by its Samples

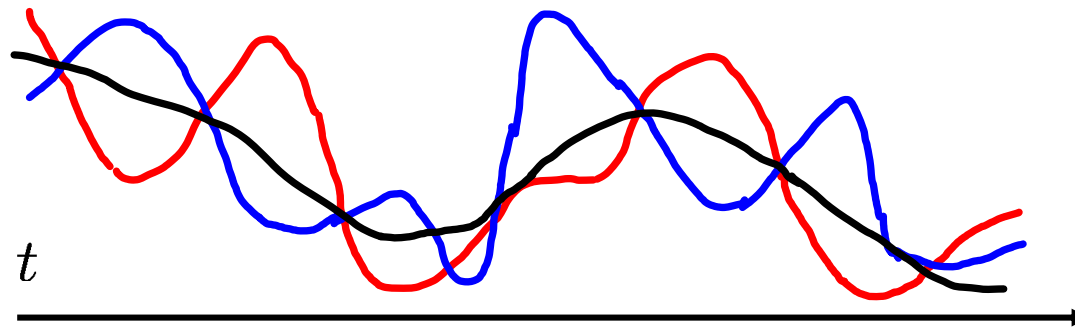


■ Representation of CT Signals by its Samples

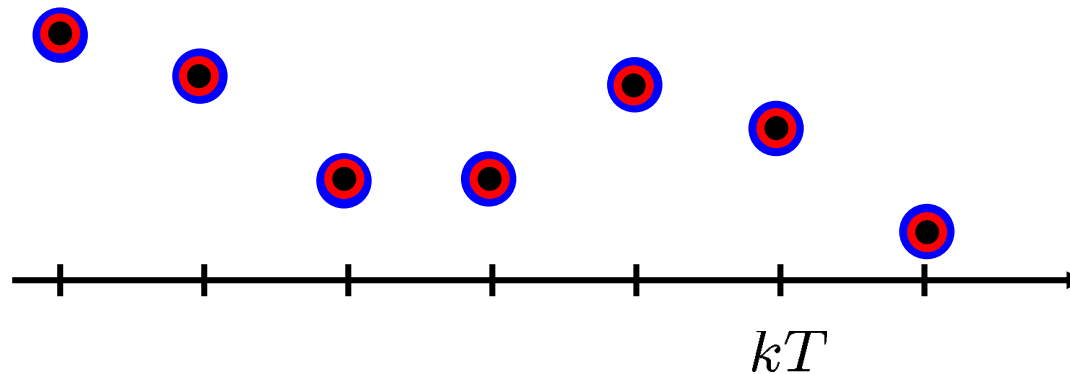


Representation of CT Signals by its Samples

$$x_1(t) \neq x_2(t) \neq x_3(t)$$



$$x_1(kT) = x_2(kT) = x_3(kT)$$



■ Impulse-Train Sampling:

$p(t)$: sampling function

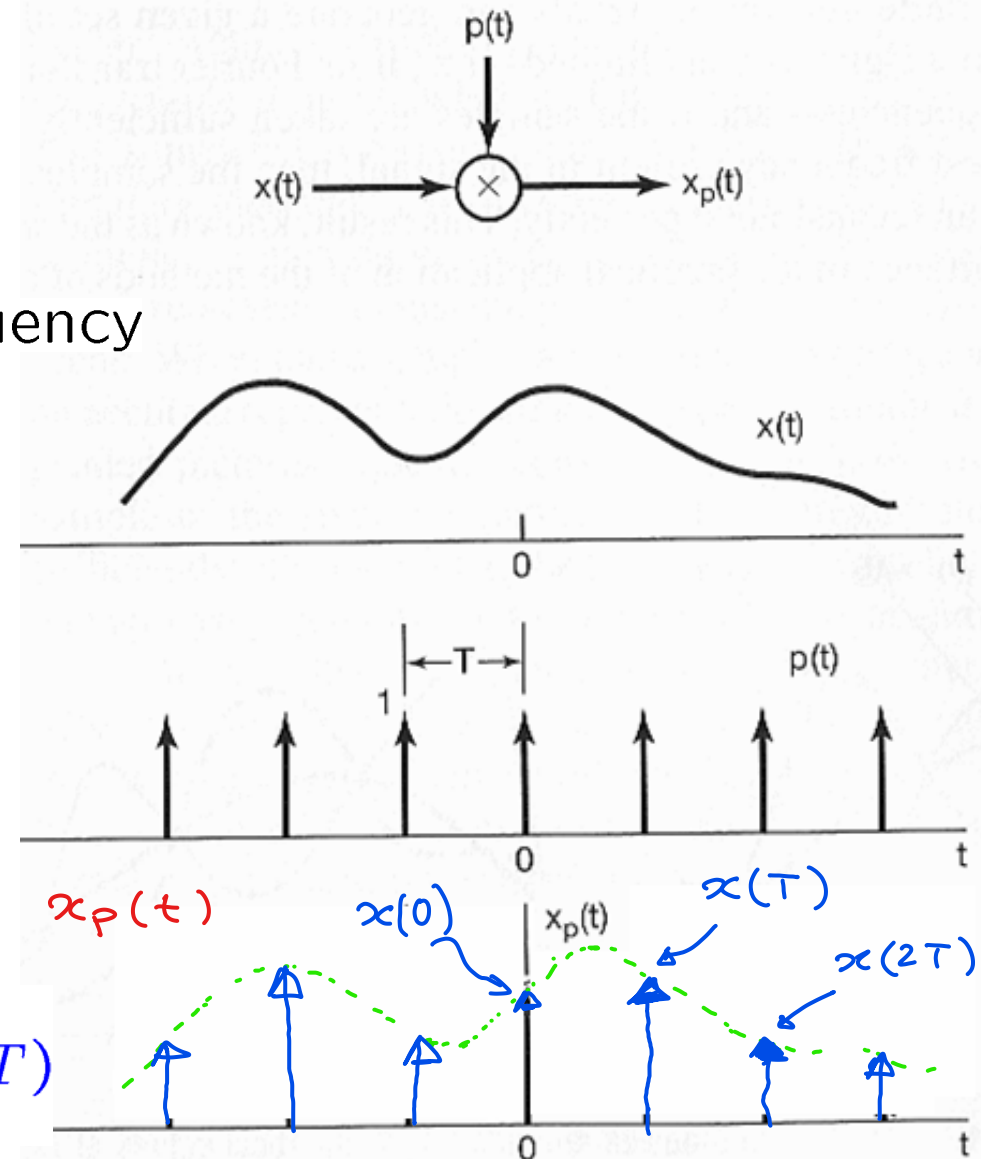
T : sampling period

$w_s = \frac{2\pi}{T}$: sampling frequency

$$\Rightarrow x_p(t) = x(t) p(t)$$

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$$

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT)$$



■ Impulse-Train Sampling:

From multiplication property,

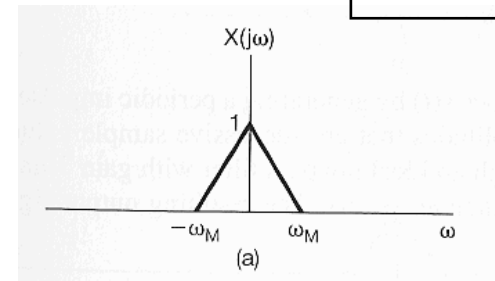
$$x_p(t) = x(t) p(t) \xleftrightarrow{\mathcal{F}} X_p(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) P(j(\omega - \theta)) d\theta$$

Eq 4.70, p. 322

$$= \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s))$$

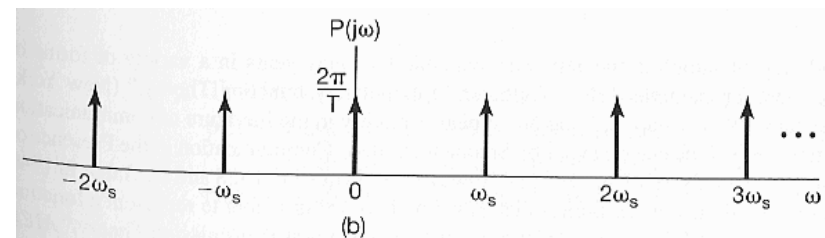
Ex 4.21, p. 323

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$



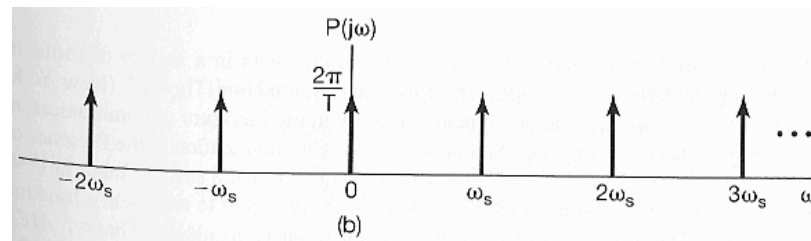
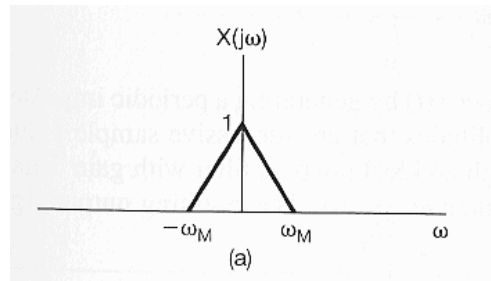
$$p(t) \xleftrightarrow{\mathcal{F}} P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

Ex 4.8, pp. 299-300

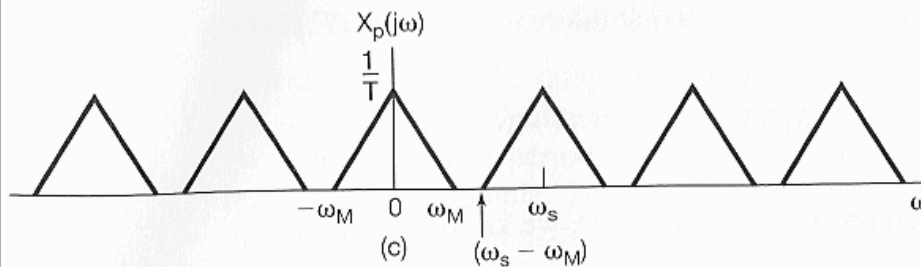


■ Impulse-Train Sampling:

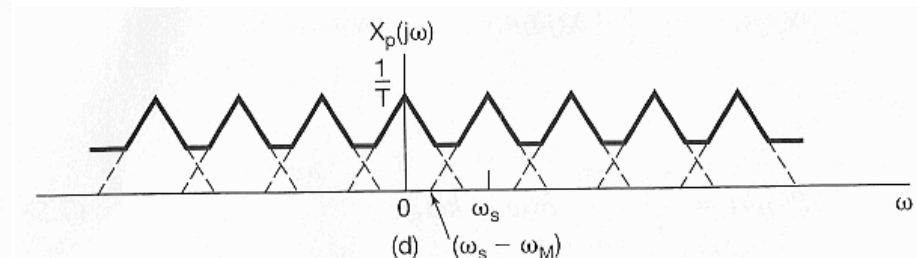
Ex 4.21, 4.22, pp. 323-4



$$\omega_s > 2\omega_M$$



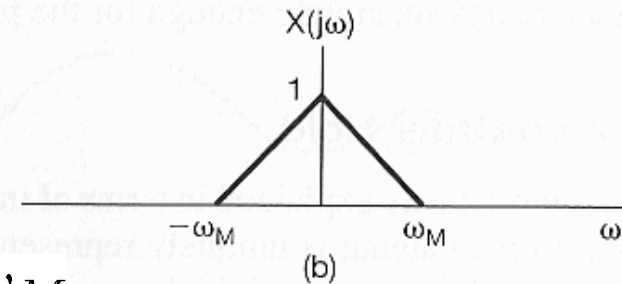
$$\omega_s < 2\omega_M$$



■ The Sampling Theorem:

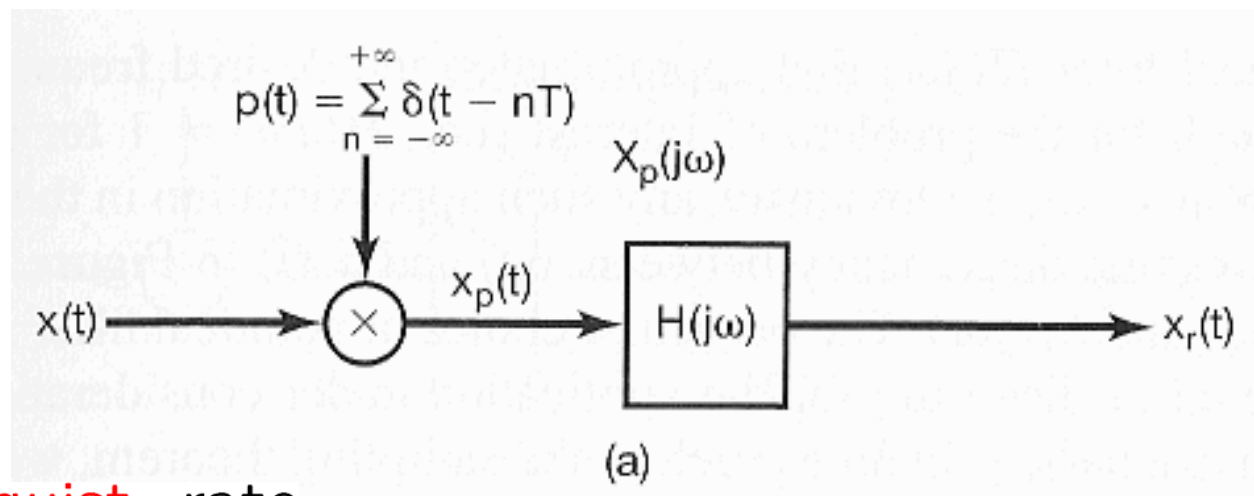
$x(t)$: a band-limited signal

with $X(j\omega) = 0$ for $|\omega| > \omega_M$



if $\omega_s > 2\omega_M$ where $\omega_s = \frac{2\pi}{T}$

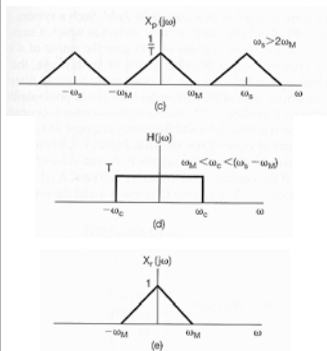
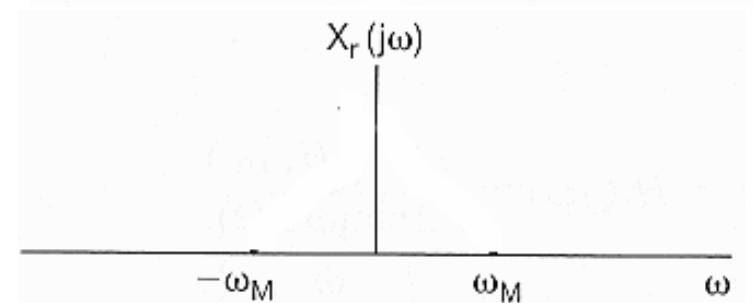
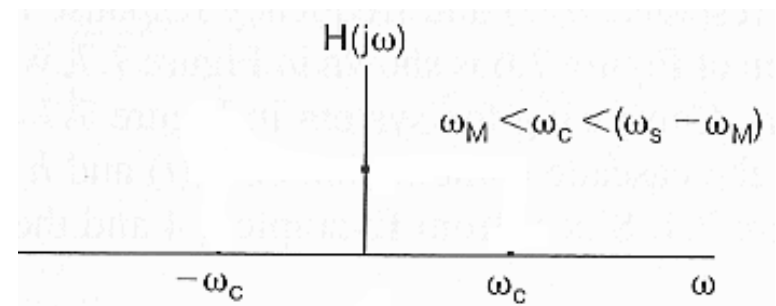
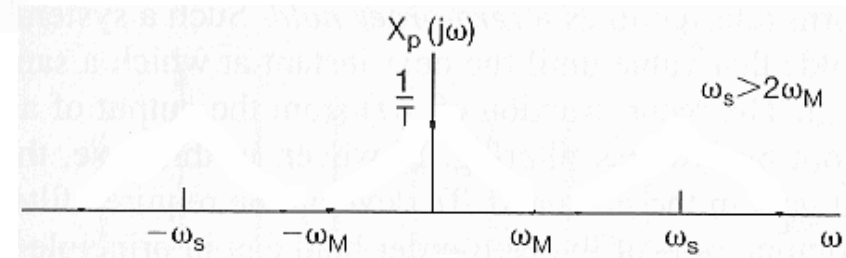
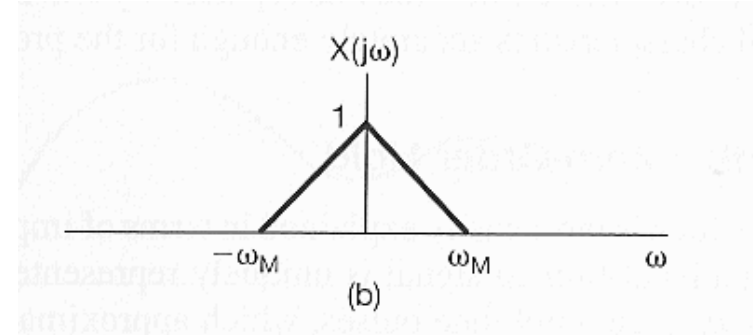
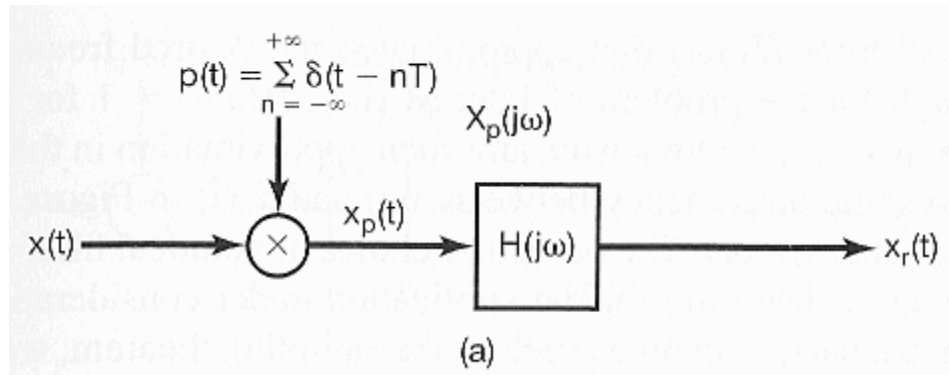
$\Rightarrow x(t)$ is uniquely determined by $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$,



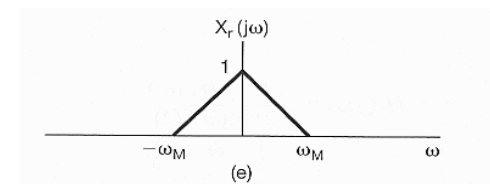
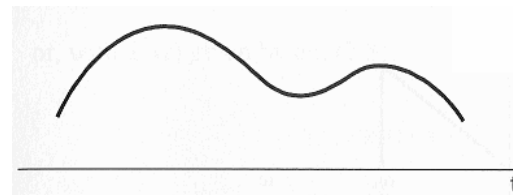
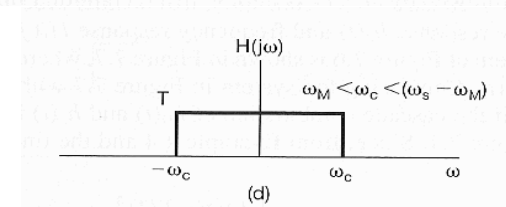
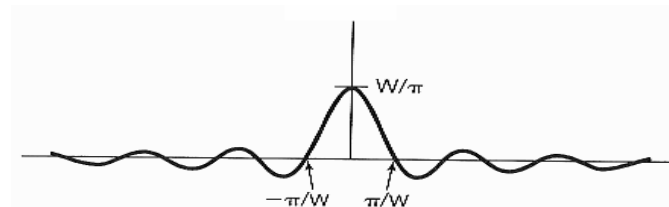
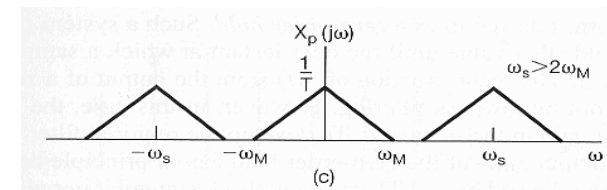
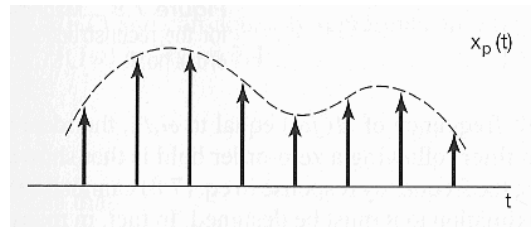
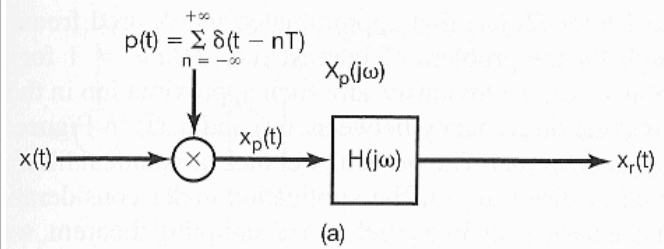
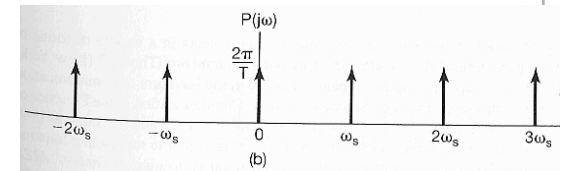
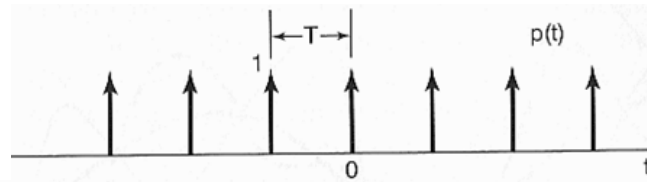
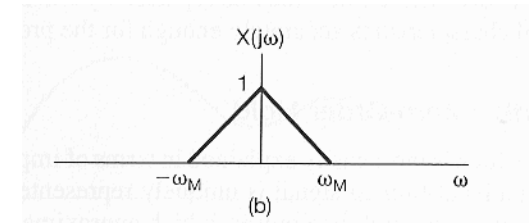
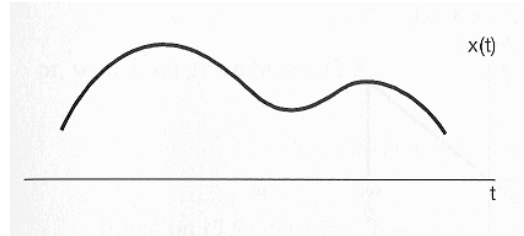
$\Rightarrow 2\omega_M$: Nyquist rate

ω_M : Nyquist frequency

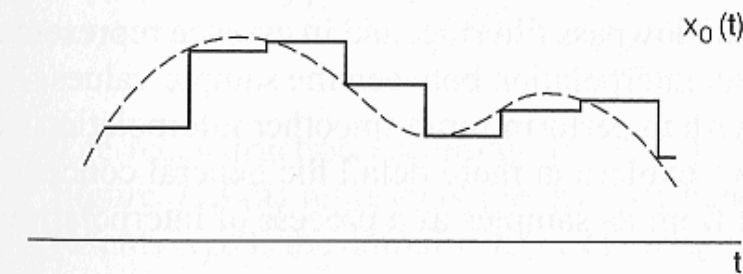
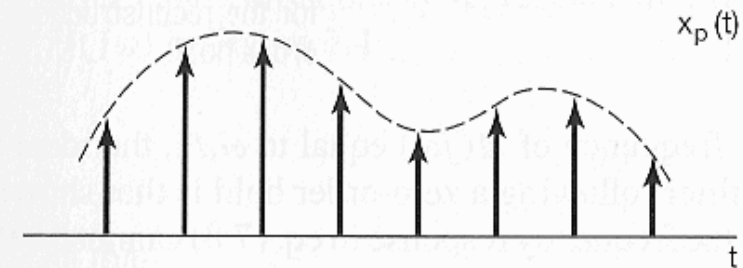
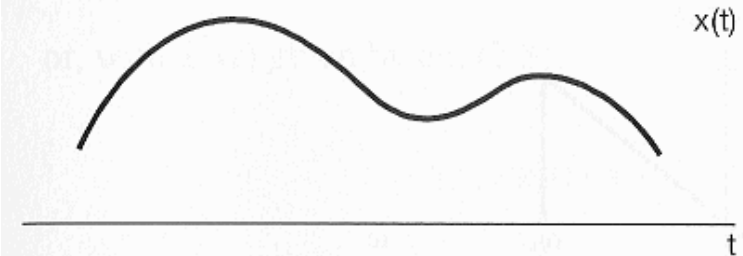
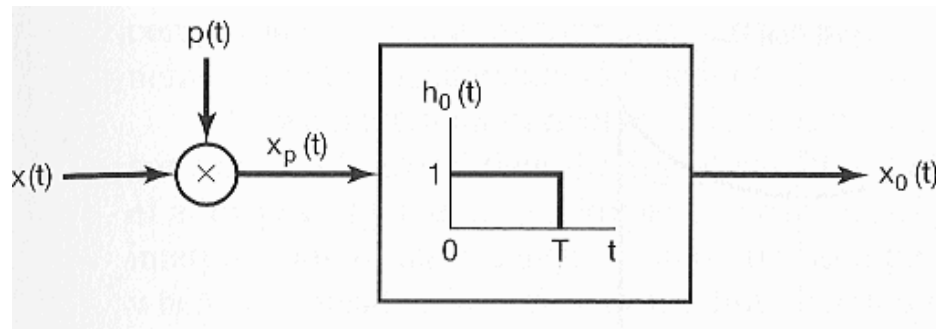
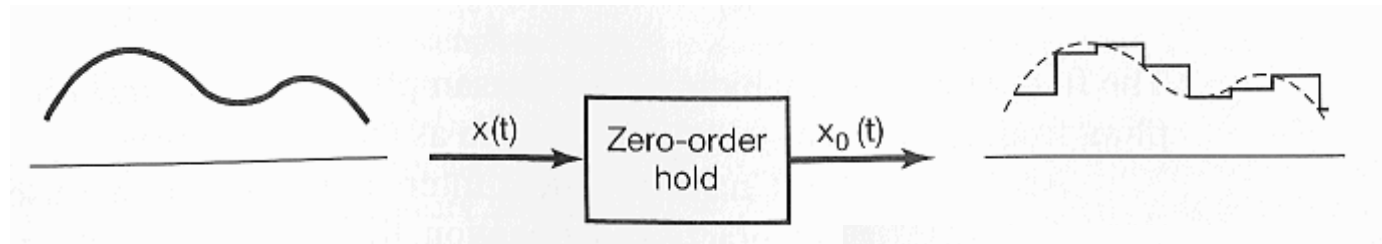
Exact Recovery by an Ideal Lowpass Filter:



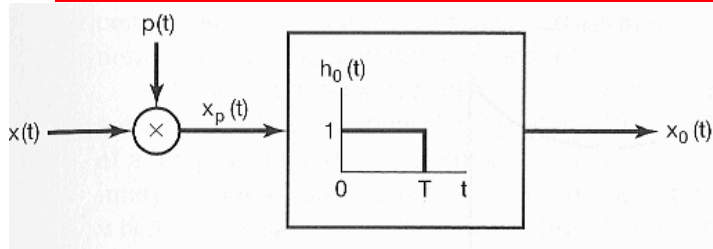
Exact Recovery by an Ideal Lowpass Filter:



■ Sampling with Zero-Order Hold:



■ Sampling with Zero-Order Hold:



Ex 4.4, p. 293

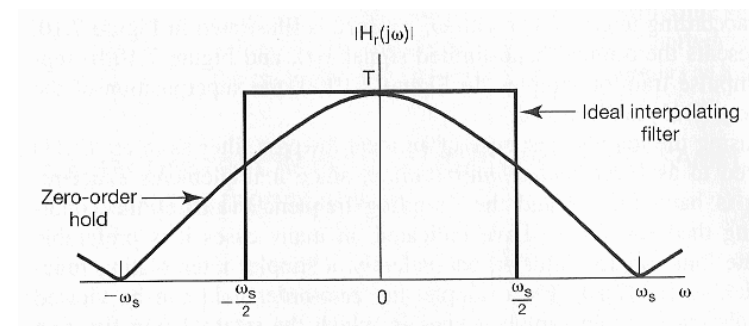
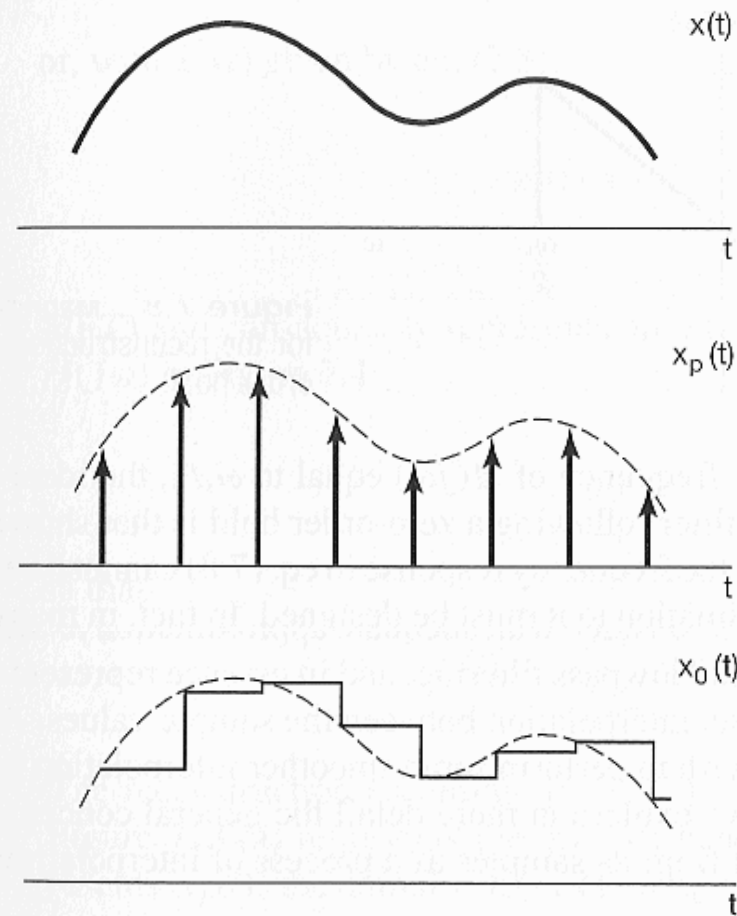
Eq 4.27, p. 301

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$

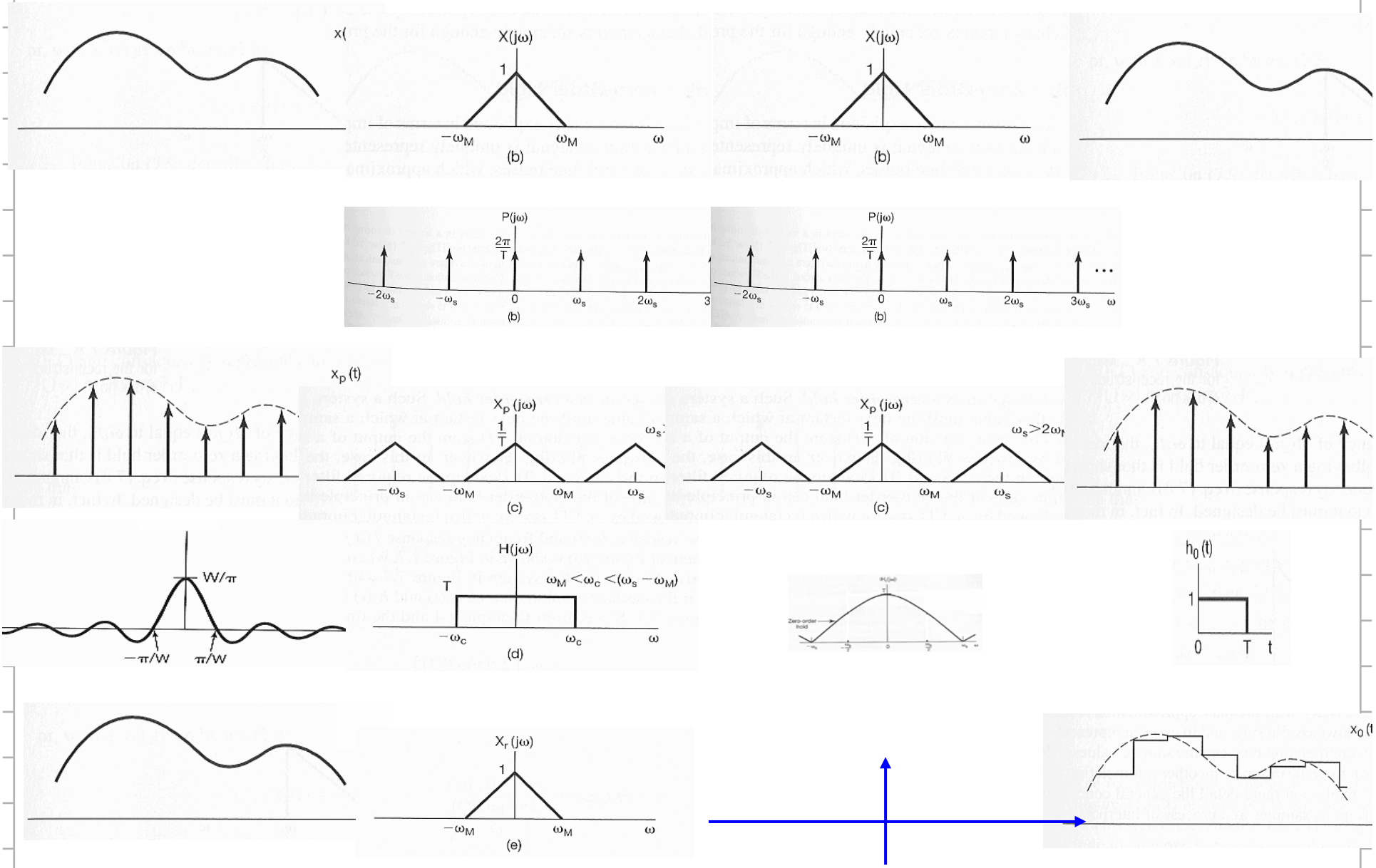
$$X(j\omega) = 2 \frac{\sin(\omega T_1)}{\omega}$$

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$$

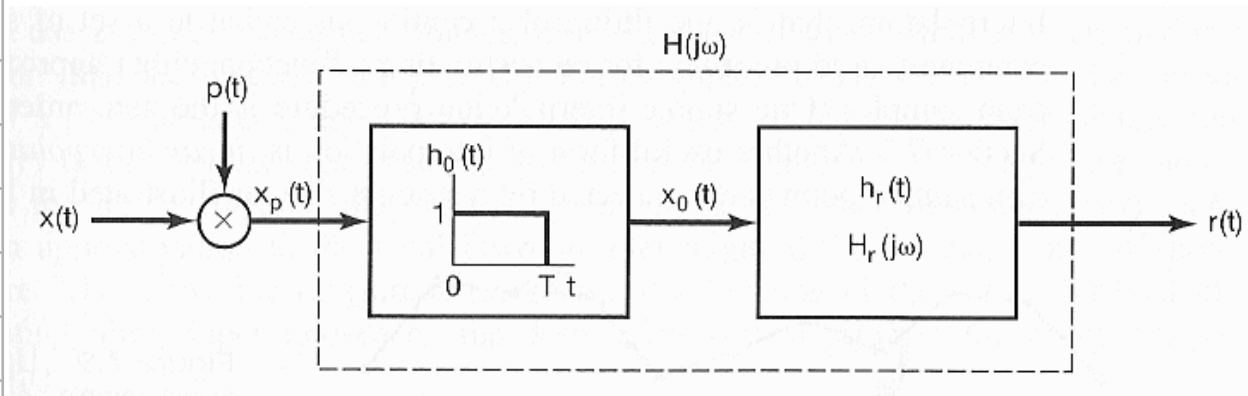
$$H_0(j\omega) = e^{-j\omega T/2} \left[\frac{2 \sin(\omega T/2)}{\omega} \right]$$



■ With Ideal Lowpass Filter & with Zero-Order Hold:



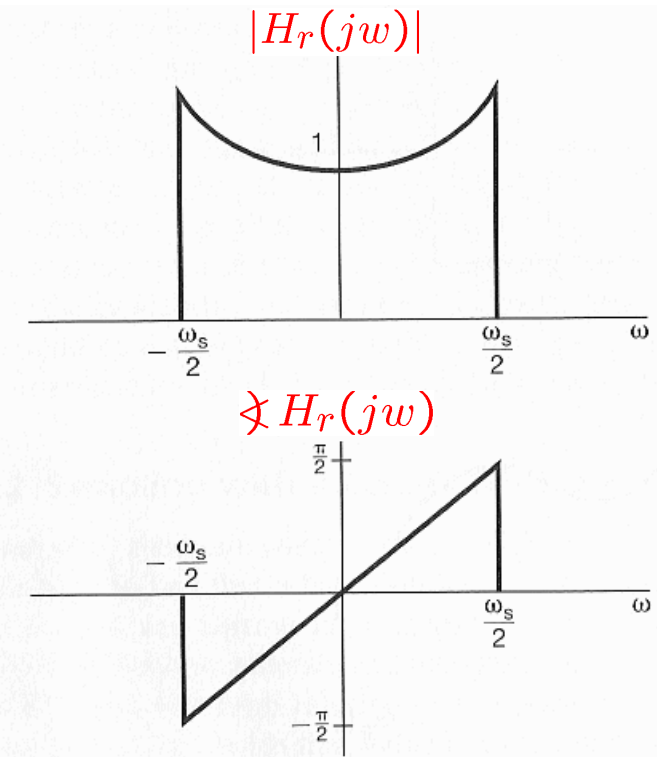
■ Sampling with Zero-Order Hold:



$$H_0(j\omega) = e^{-j\omega T/2} \left[\frac{2 \sin(\omega T/2)}{\omega} \right]$$

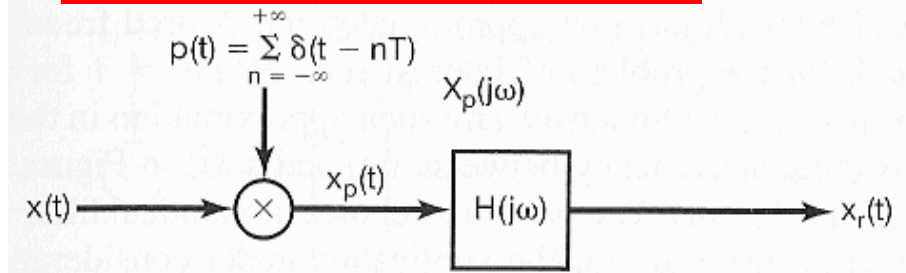
$$H(j\omega) = H_0(j\omega) H_r(j\omega)$$

$$\Rightarrow H_r(j\omega) = \frac{e^{j\omega T/2} H(j\omega)}{\frac{2 \sin(\omega T/2)}{\omega}}$$



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Exact Interpolation:



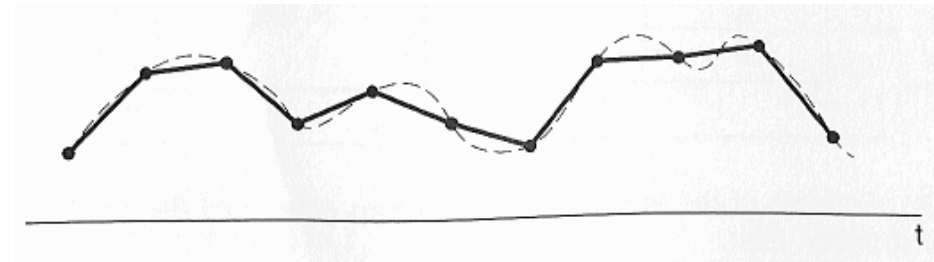
$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT)$$

$$x_r(t) = x_p(t) * h(t)$$

Ex 2.11, p. 110 $x(t - t_0) = x(t) * \delta(t - t_0)$

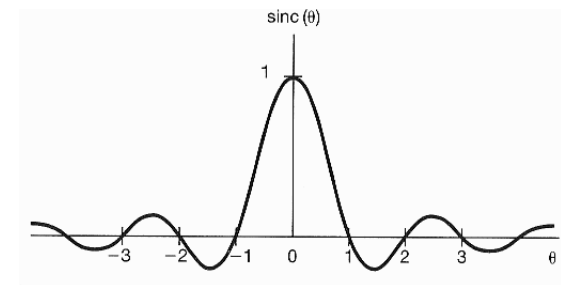
$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT) h(t - nT)$$

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT) \frac{w_c T}{\pi} \frac{\sin(w_c(t - nT))}{w_c(t - nT)}$$

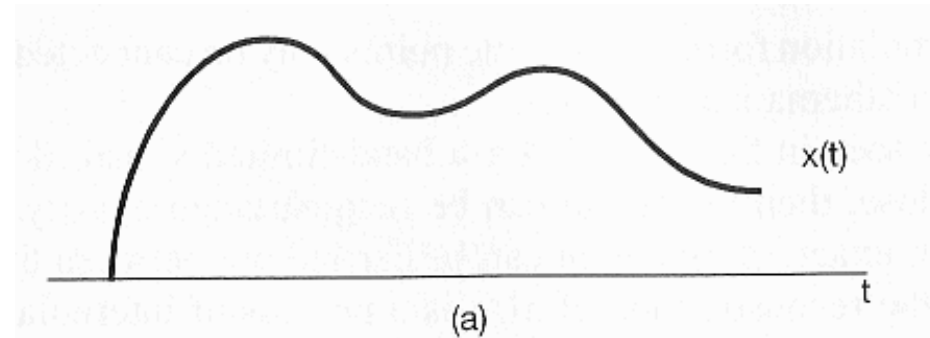
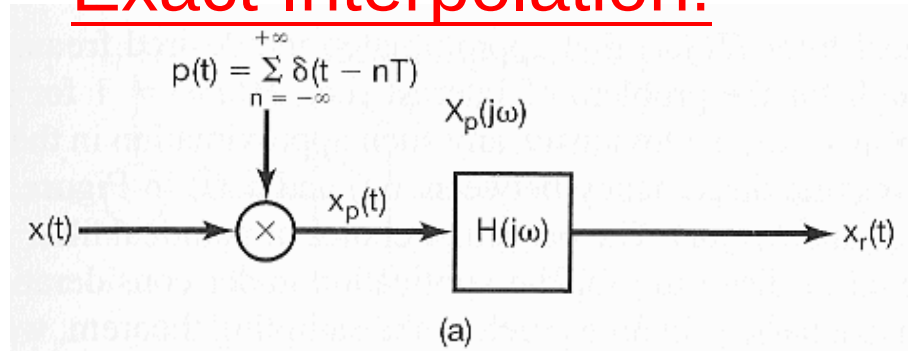


ideal lowpass filter
with a magnitude of T

$$h(t) = T \frac{w_c}{\pi} \frac{\sin(w_c t)}{w_c t}$$



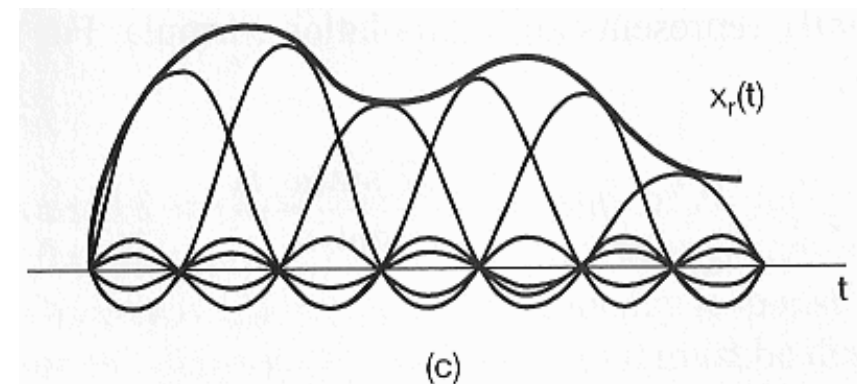
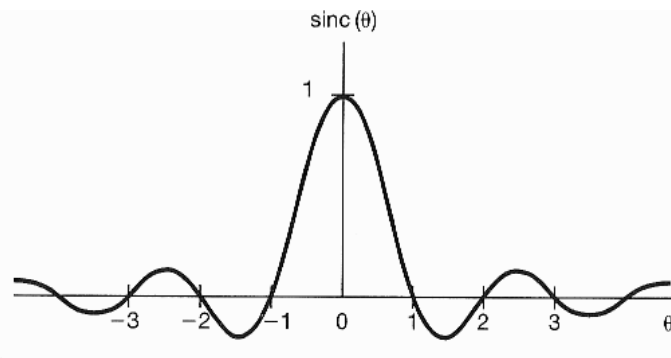
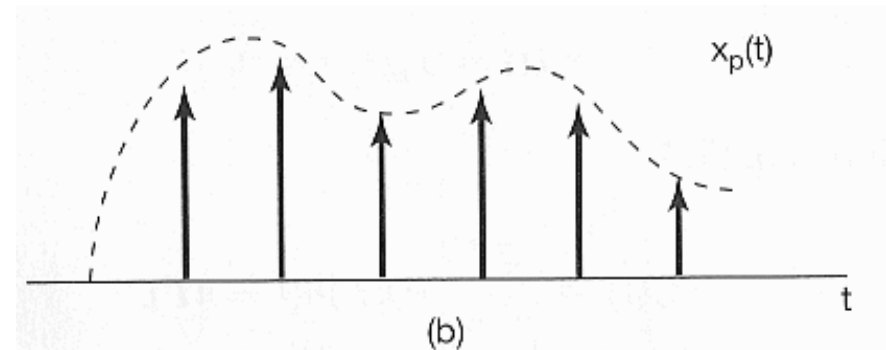
Exact Interpolation:



$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT) \frac{\omega_c T}{\pi} \frac{\sin(\omega_c(t - nT))}{\omega_c(t - nT)}$$

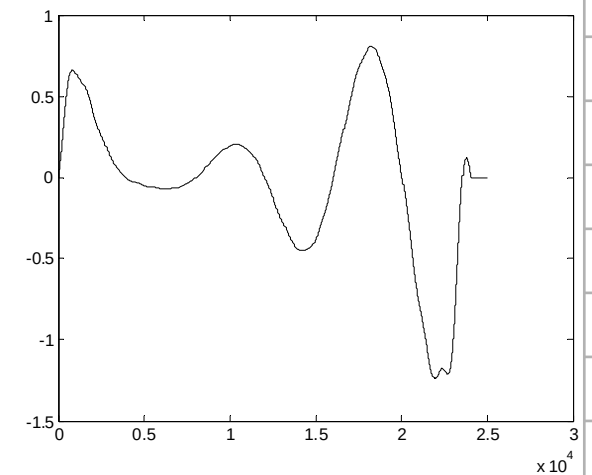
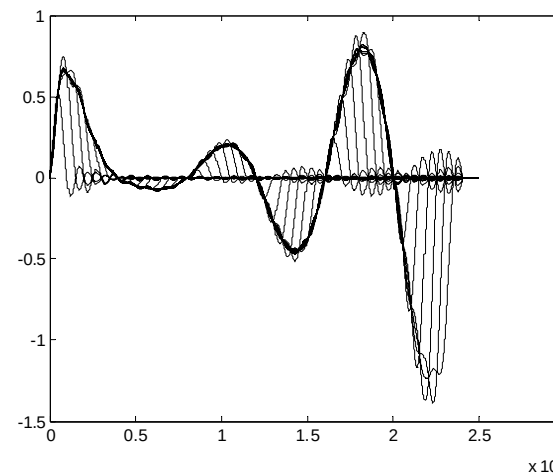
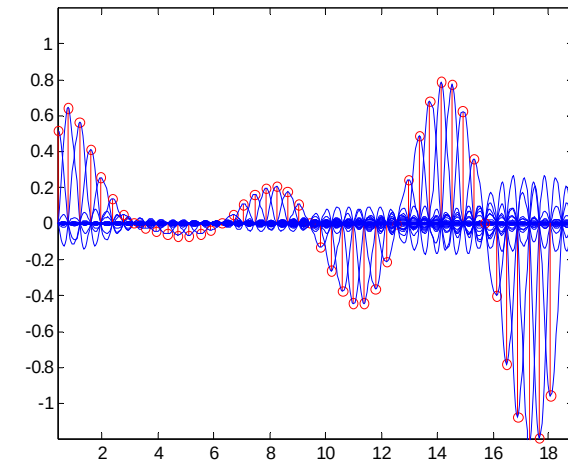
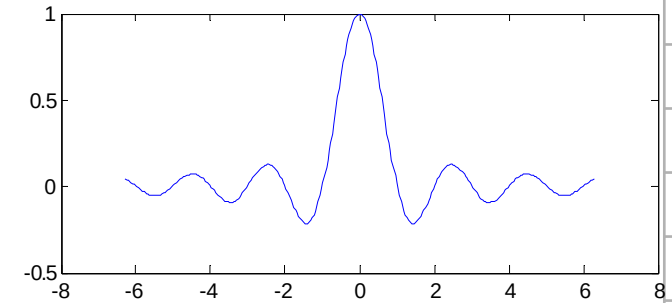
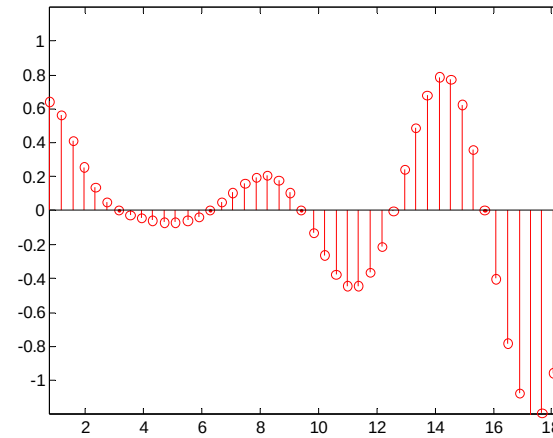
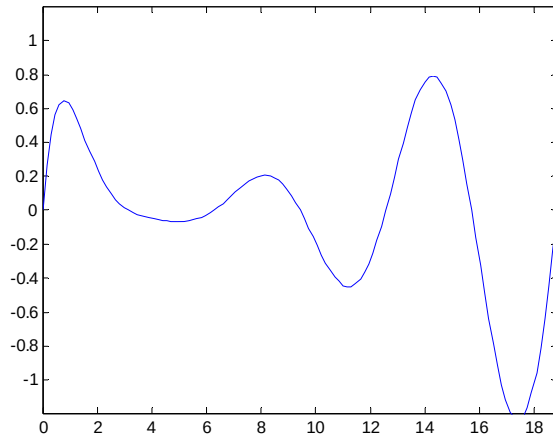
$$\frac{\omega_c}{\pi} \frac{2\pi}{\omega_s} \frac{\sin \pi(\omega_c(t - nT)/\pi)}{\pi \omega_c(t - nT)/\pi}$$

$$\frac{2\omega_c}{\omega_s} \text{sinc}(\omega_c(t - nT)/\pi)$$

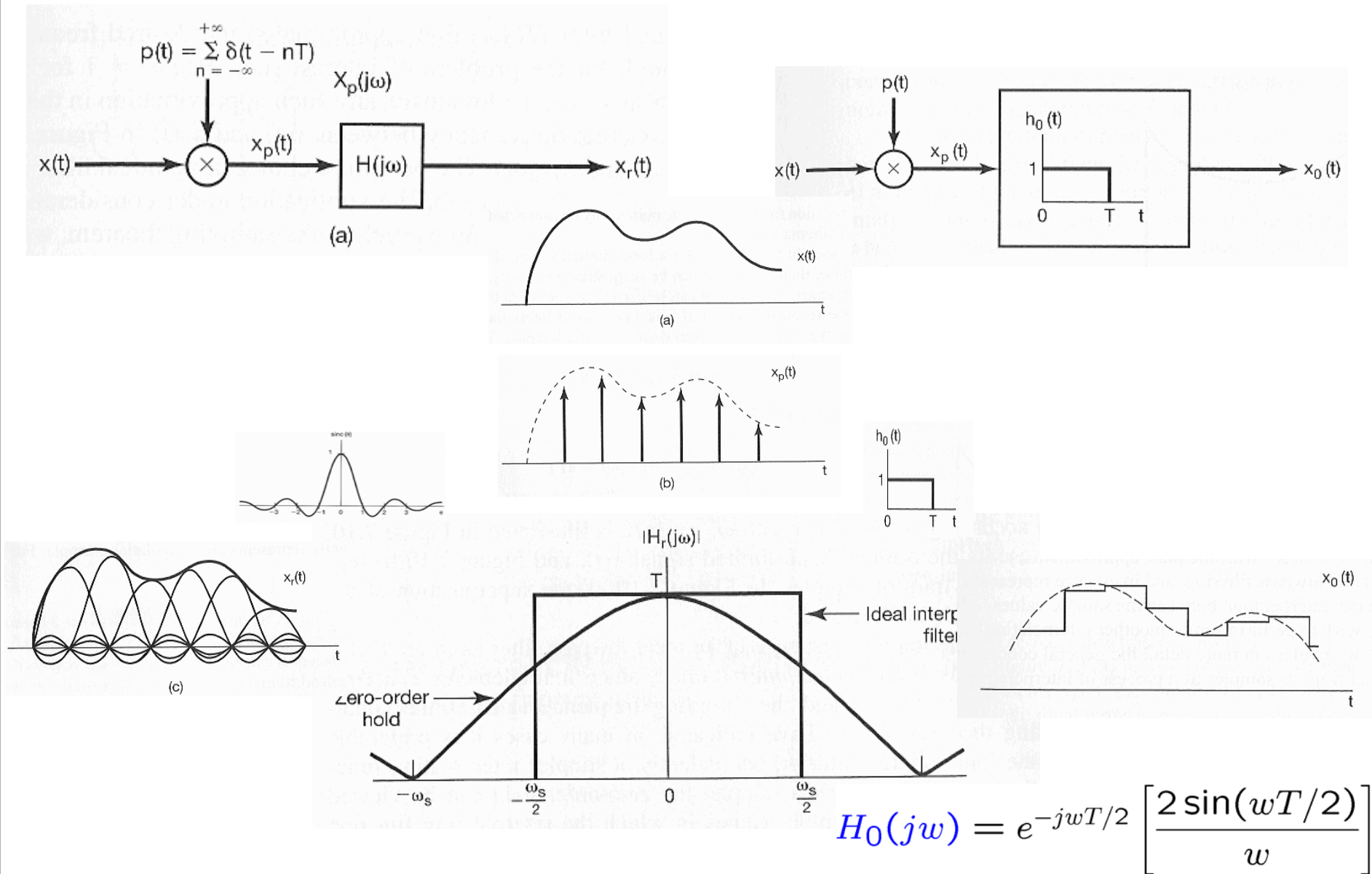


Reconstruction of a Signal from its Samples Using Interpolation

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NTU EE-SS7-Sampling-21



■ Ideal Interpolating Filter & The Zero-Order Hold:



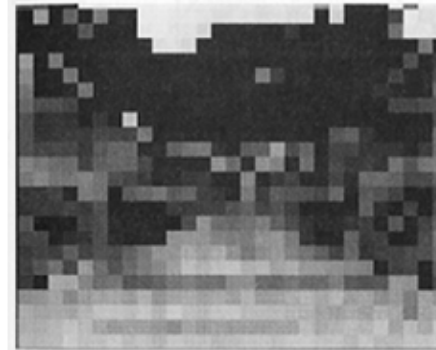
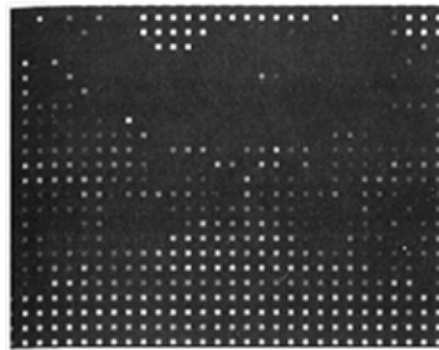
■ Sampling & Interpolation of Images:

original image impulse sampling zero-order hold zero-order hold

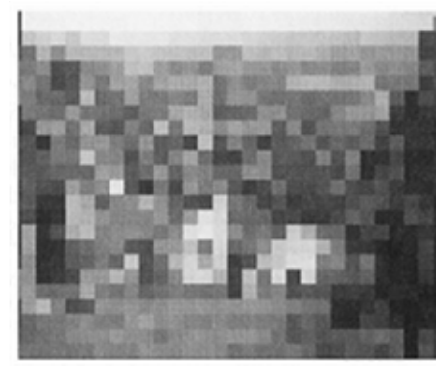
4 : 1



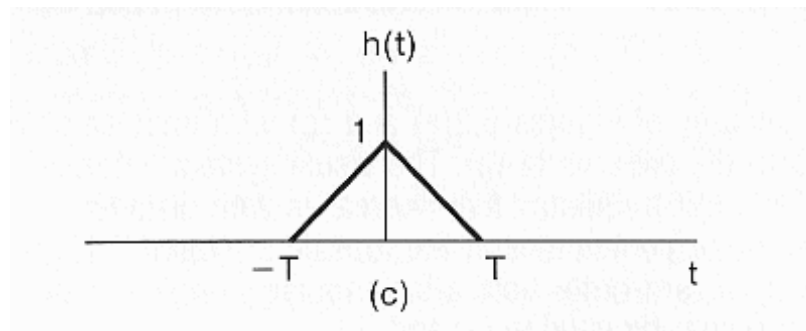
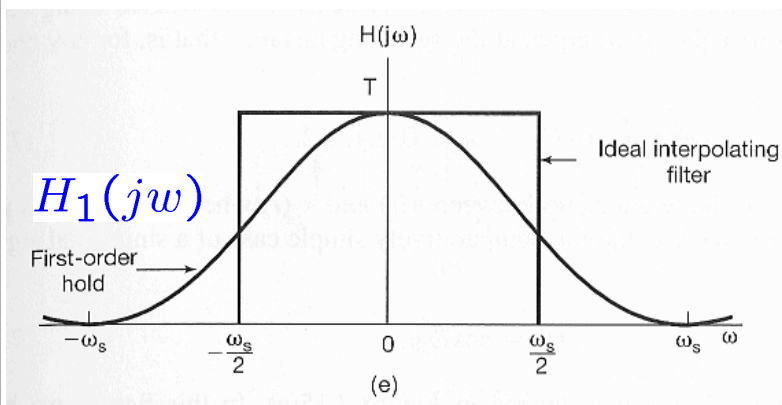
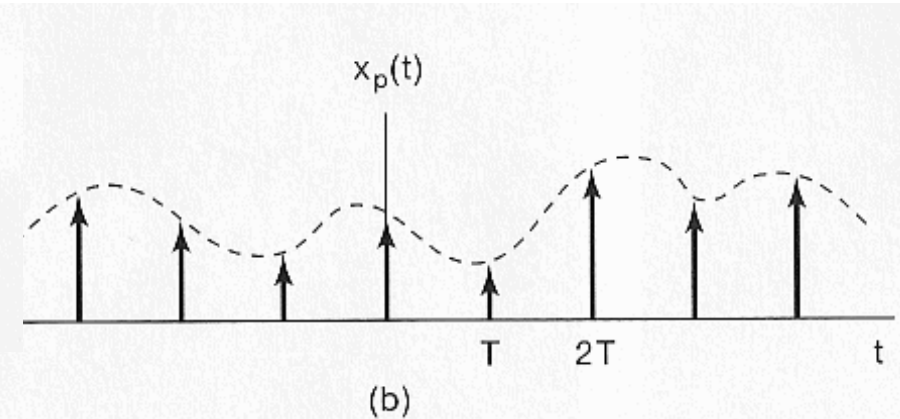
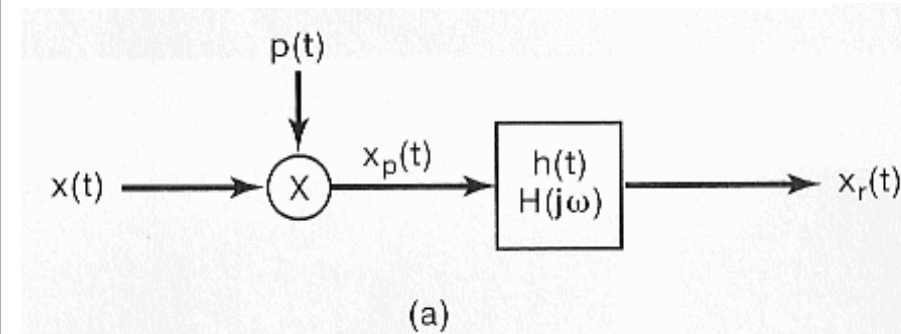
(a)



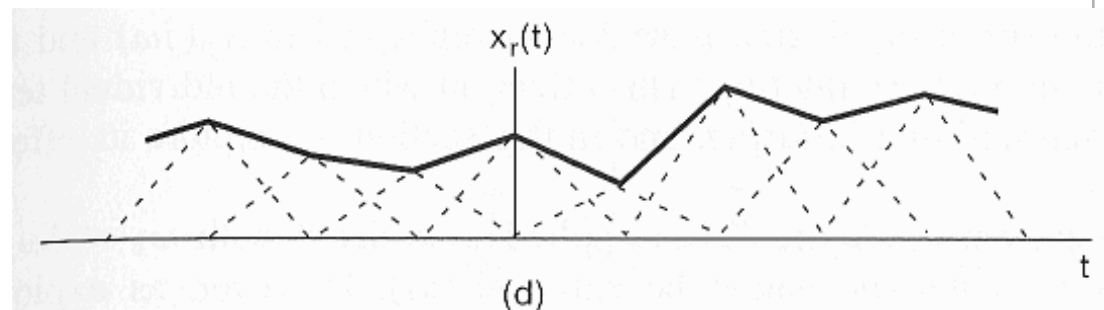
(g)



Higher-Order Holds:



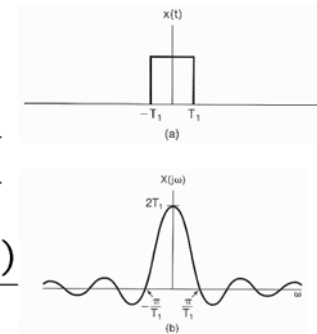
$$H_1(j\omega) = \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$



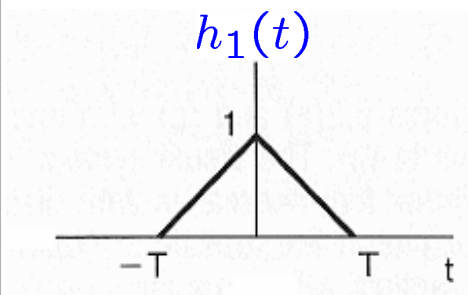
Higher-Order Holds:

Ex 4.4, p. 293

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$



$$X(j\omega) = 2 \frac{\sin(\omega T_1)}{\omega}$$

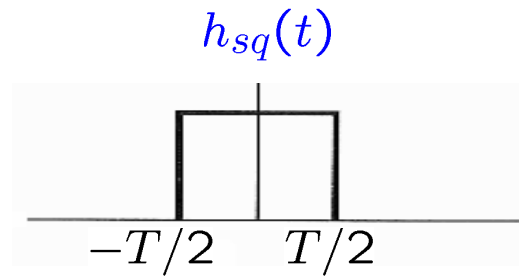


$$H_1(j\omega)$$

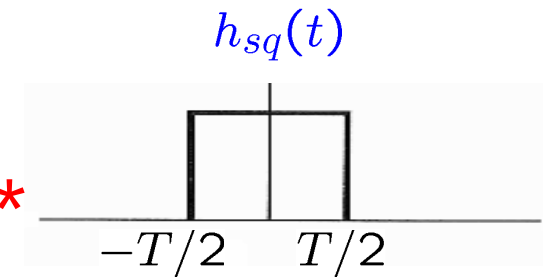
$$= \frac{1}{T}$$

$$= \frac{1}{T}$$

$$= \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$



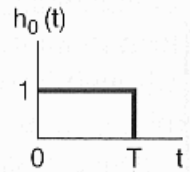
*



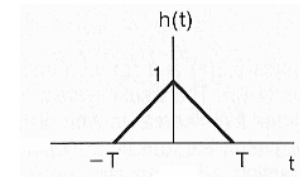
X

$$2 \frac{\sin(\omega T/2)}{\omega}$$

■ First-Order Hold on Image Processing:



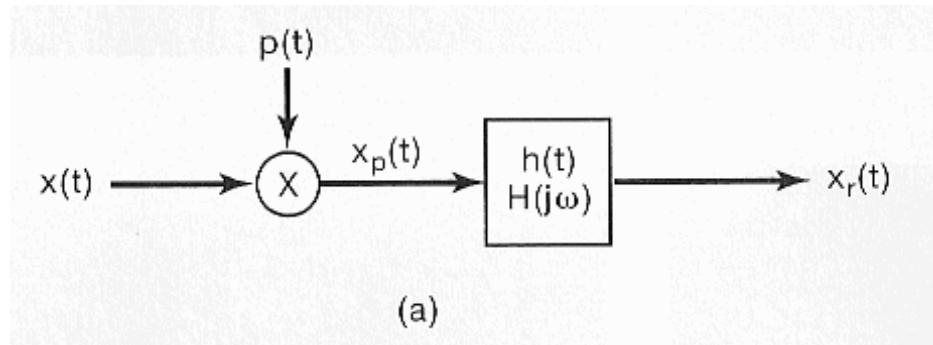
zero-order hold



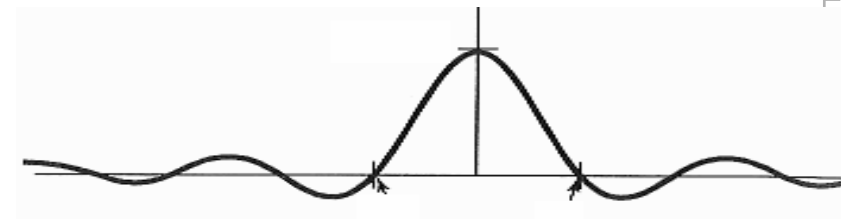
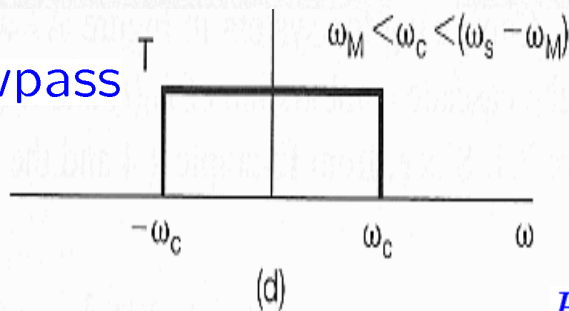
first-order hold



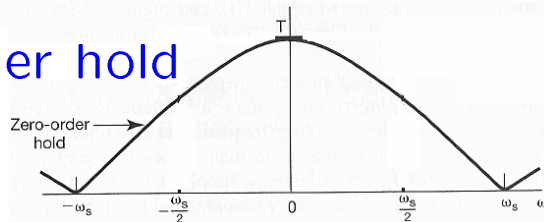
Three Filters: Ideal Lowpass, Zero-Order, First-Order



ideal lowpass

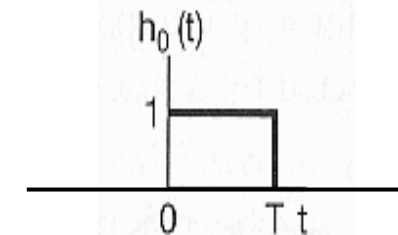


zero-order hold

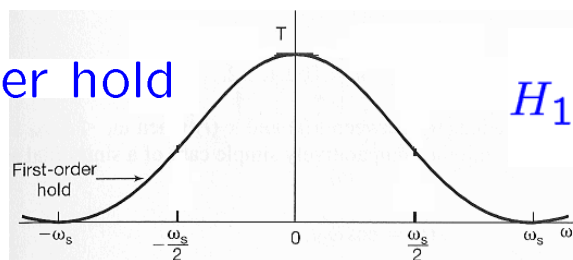


$$H_0(j\omega) =$$

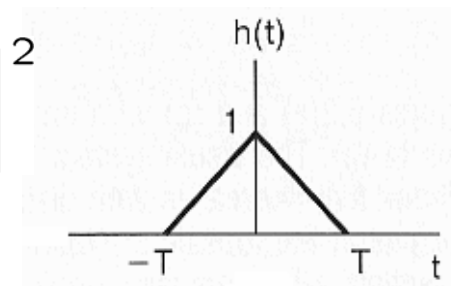
$$e^{-j\omega T/2} \left[\frac{2 \sin(\omega T/2)}{\omega} \right]$$



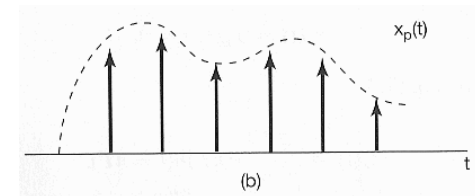
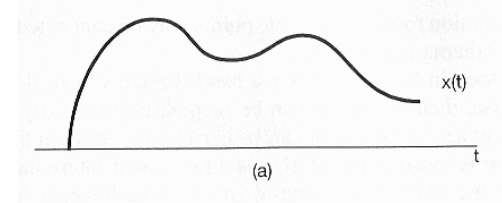
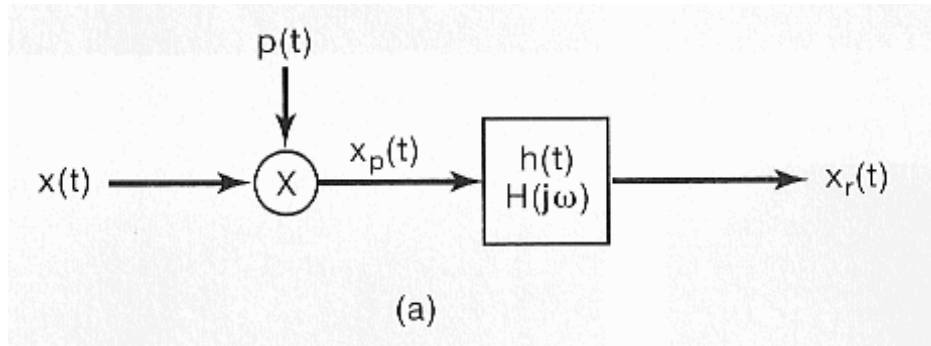
first-order hold



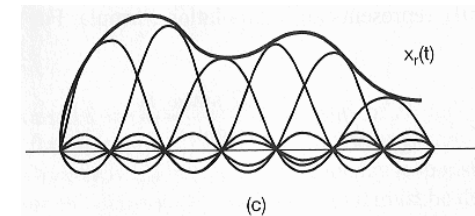
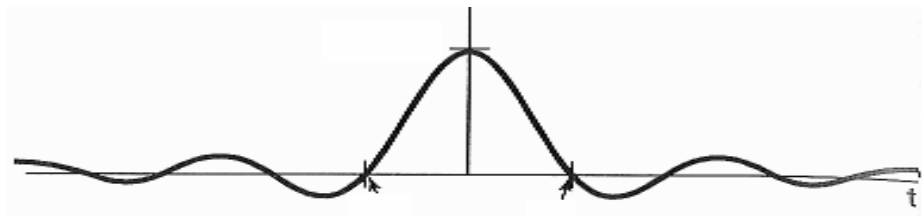
$$H_1(j\omega) = \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$



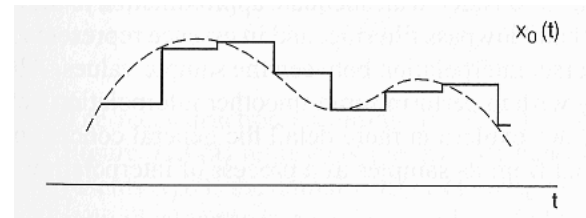
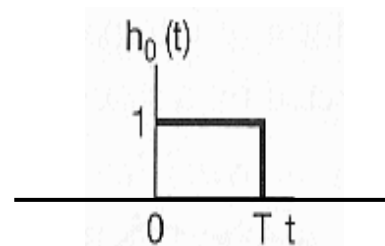
Three Filters: Ideal Lowpass, Zero-Order, First-Order



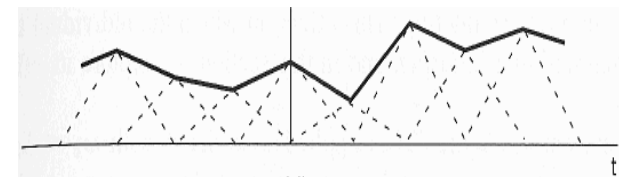
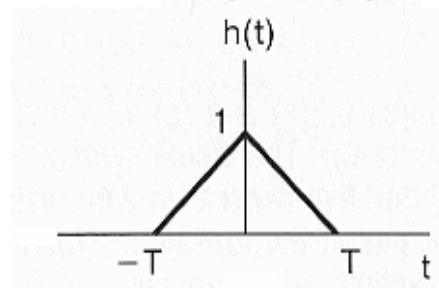
ideal lowpass



zero-order hold

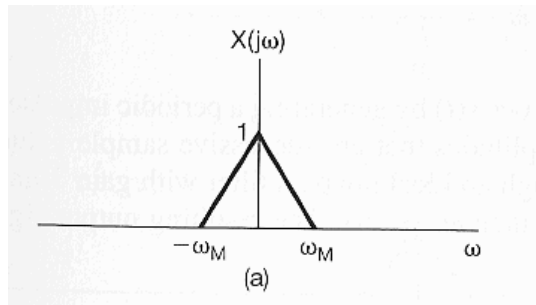


first-order hold

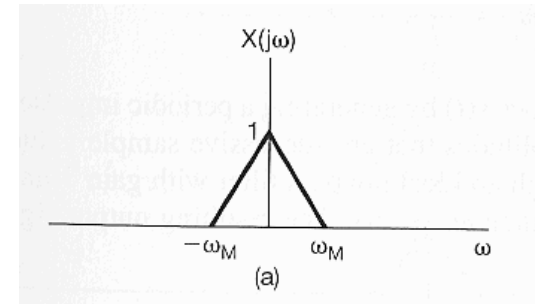


- Representation of of a Continuous-Time Signal by Its Samples: The Sampling Theorem
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- Discrete-Time Processing of Continuous-Time Signals
- Sampling of Discrete-Time Signals

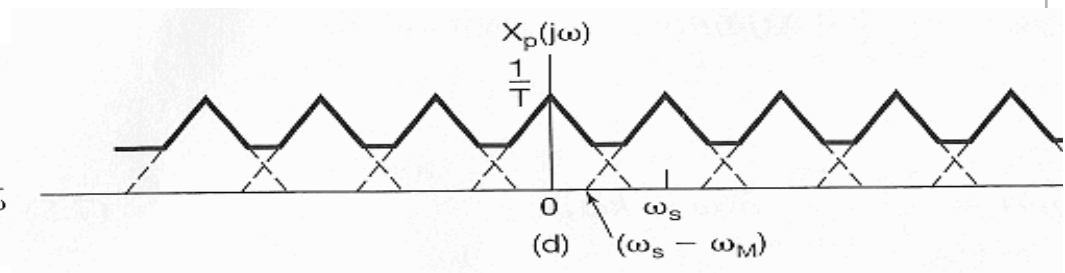
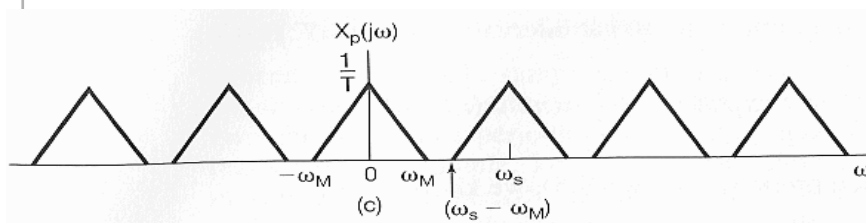
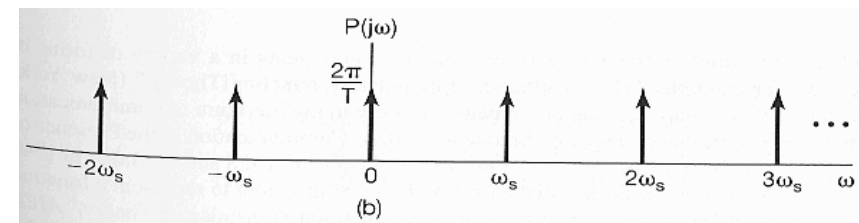
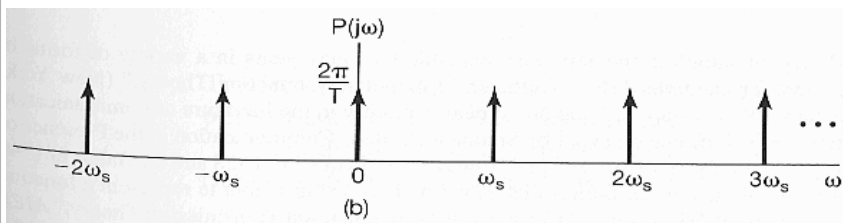
Overlapping in Frequency-Domain: Aliasing



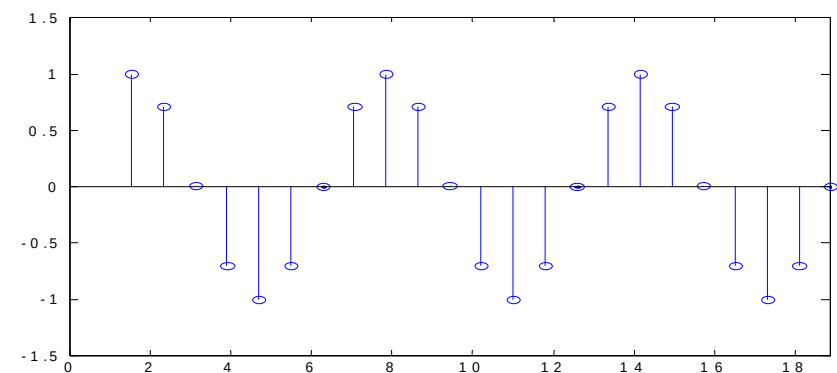
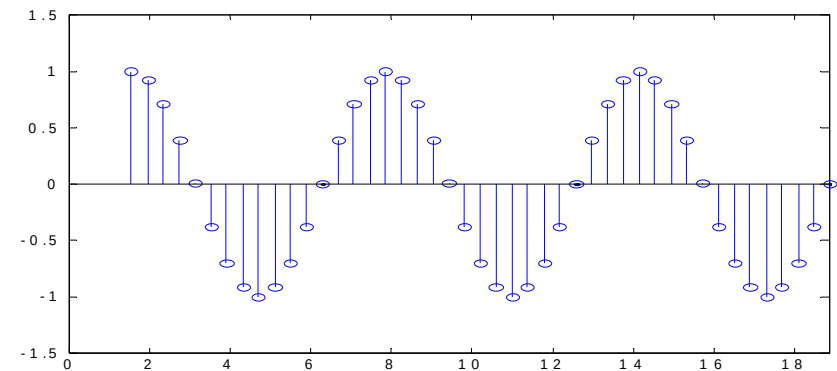
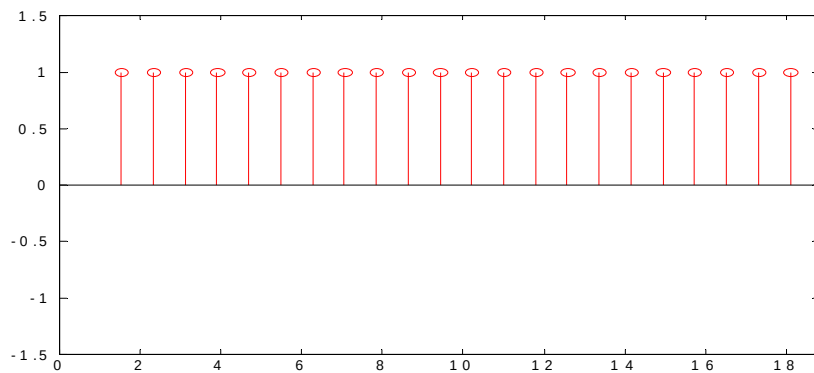
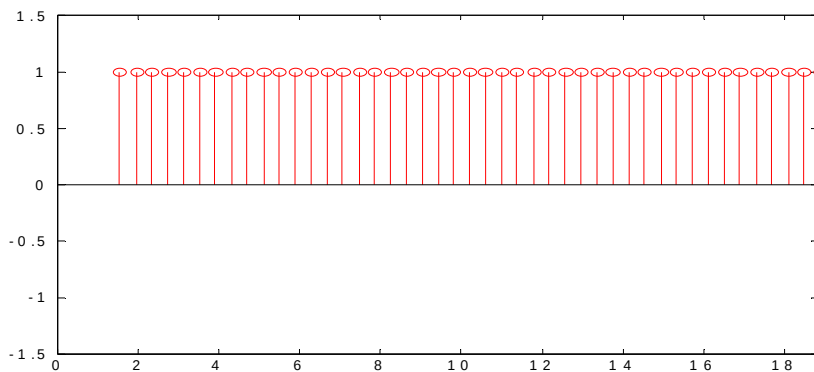
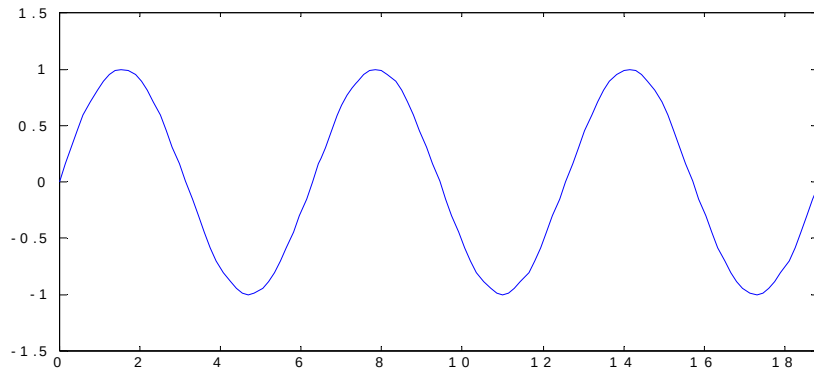
$$\omega_s > 2\omega_M$$



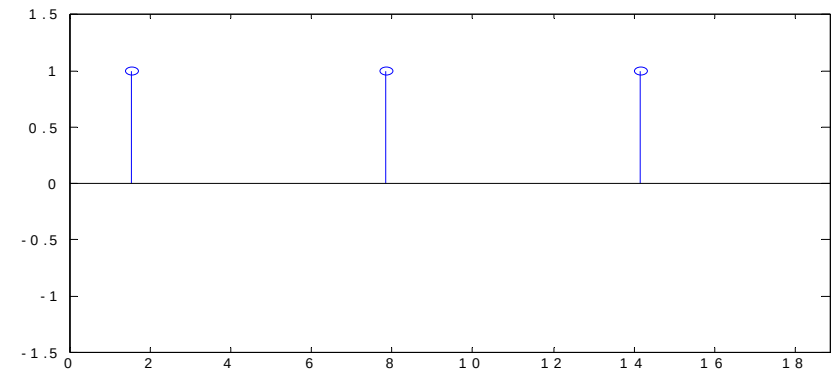
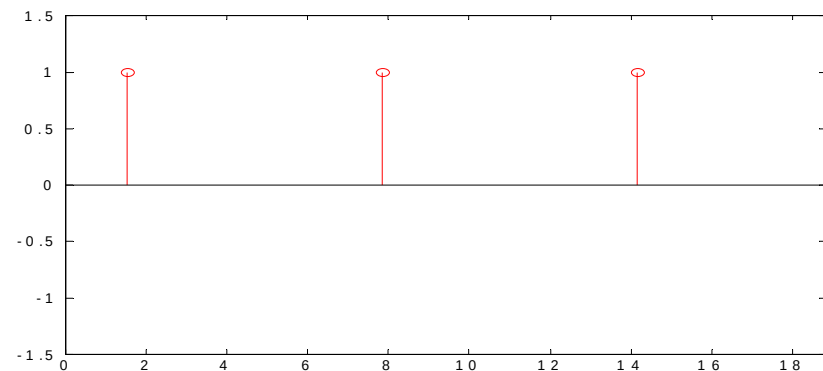
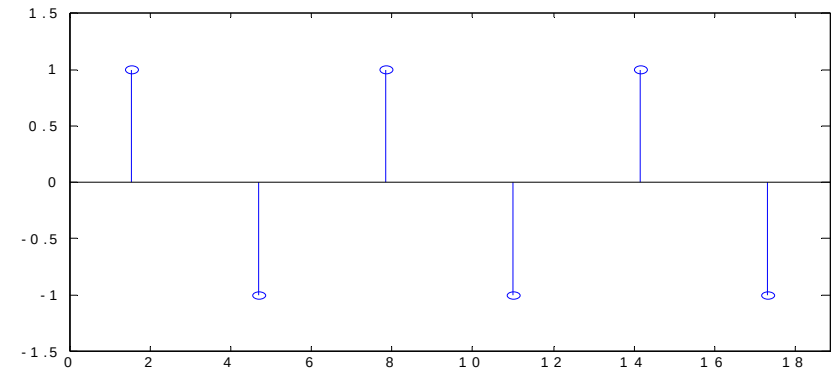
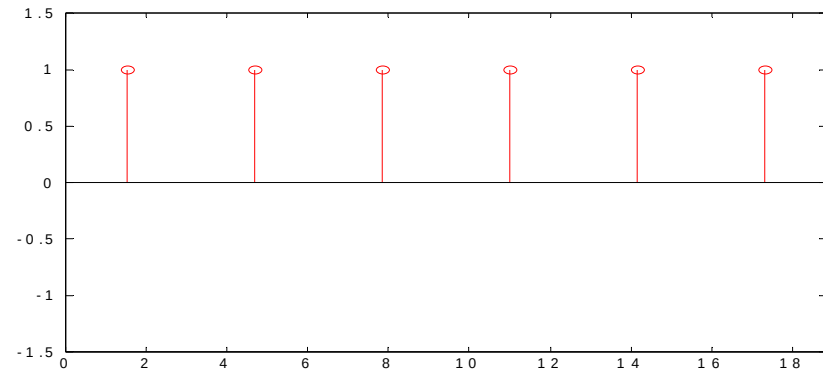
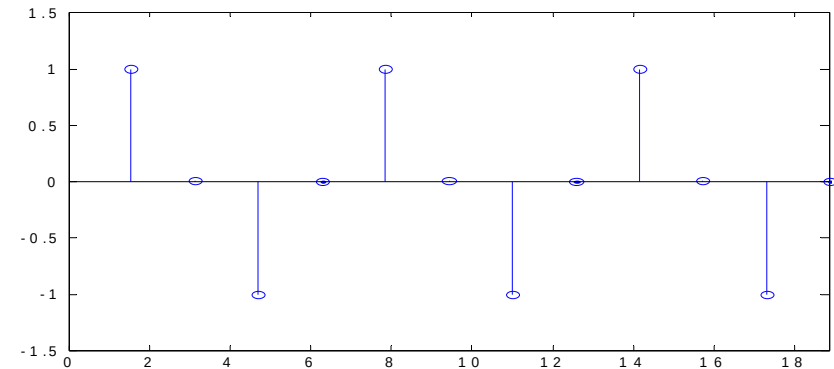
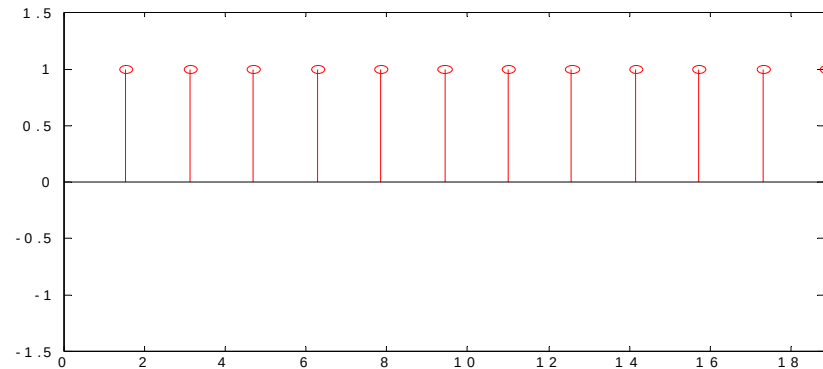
$$\omega_s < 2\omega_M$$



■ Overlapping in Frequency-Domain: Aliasing

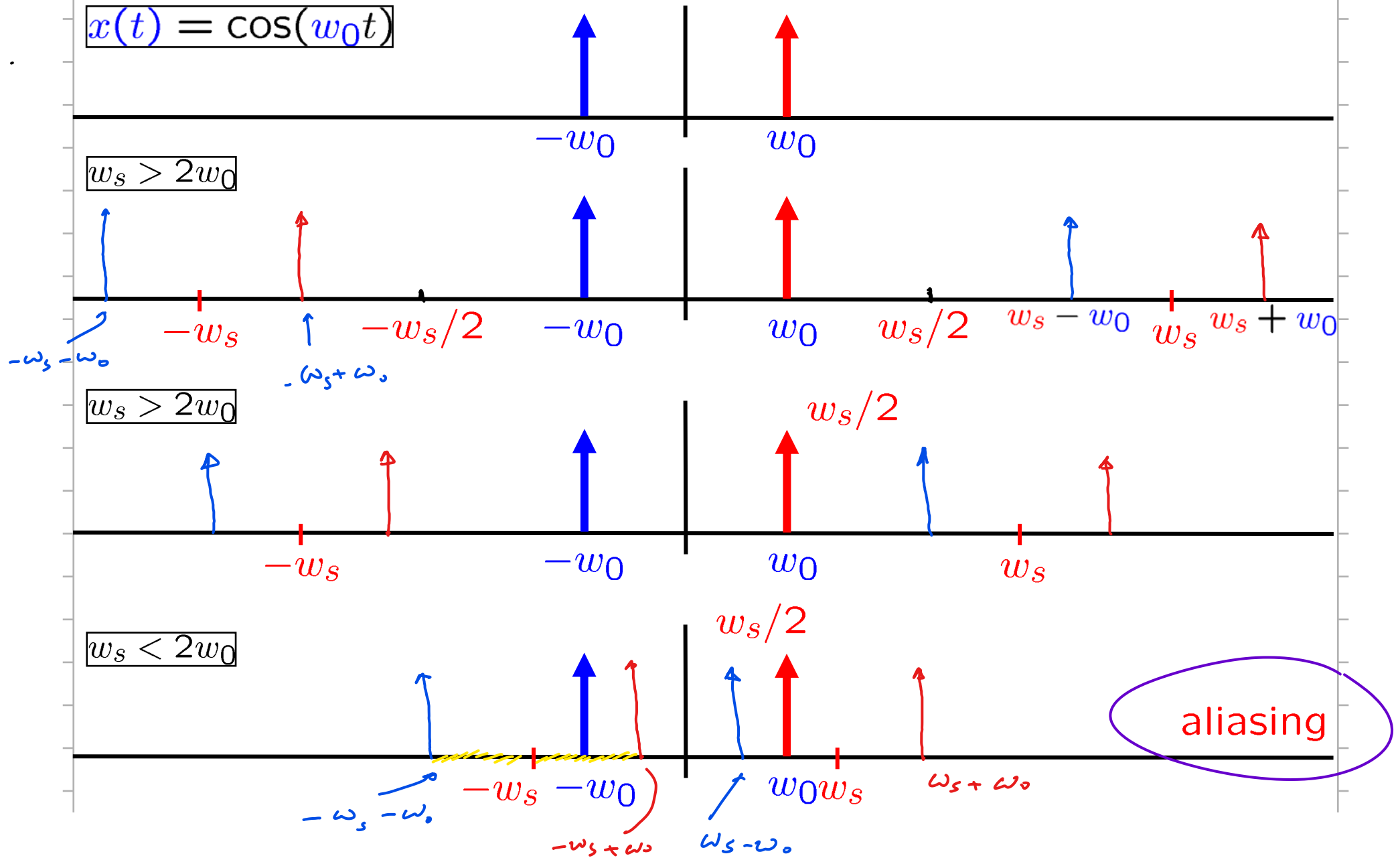


■ Overlapping in Frequency-Domain: Aliasing



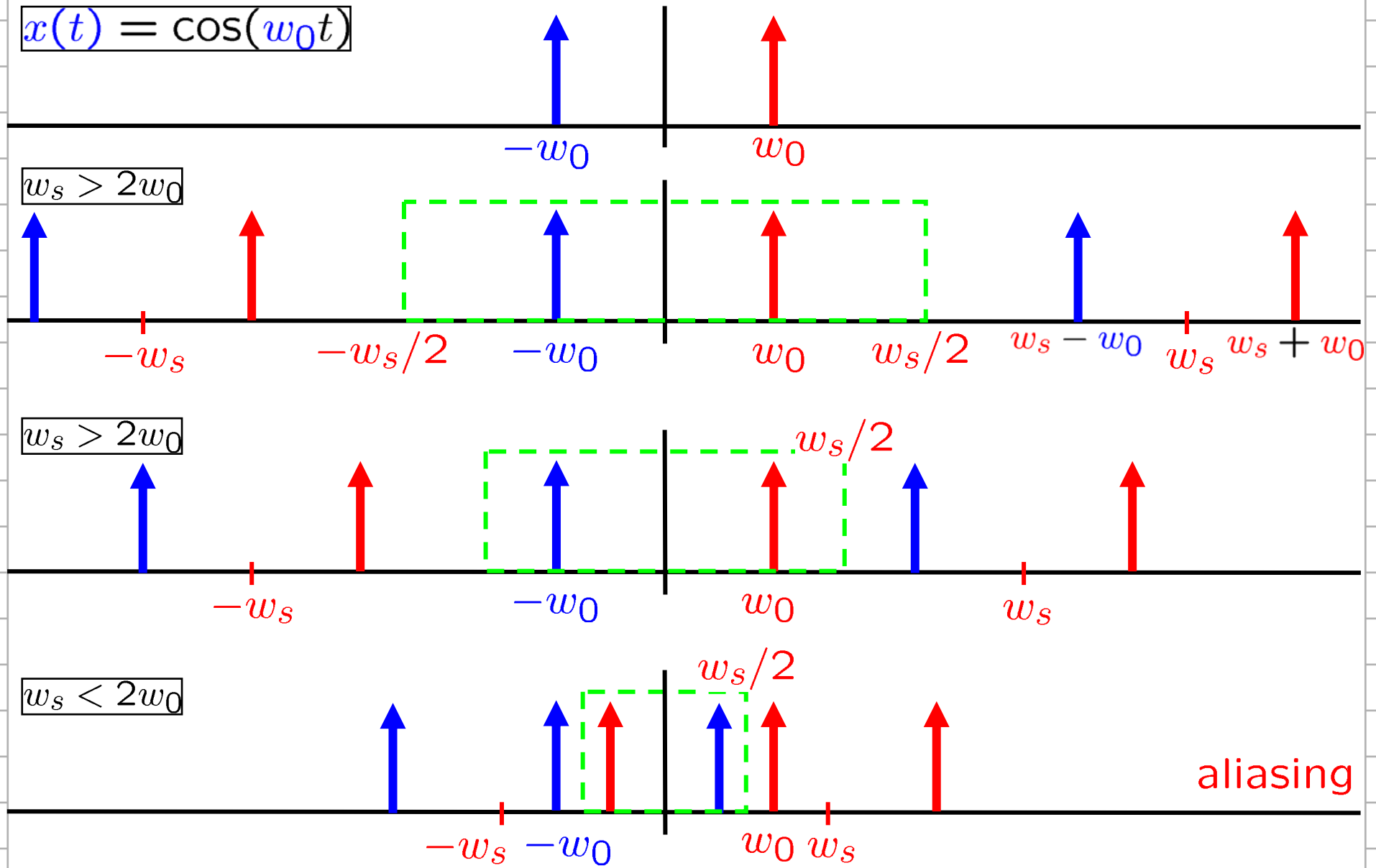
Overlapping in Frequency-Domain: Aliasing

$$x(t) = \cos(\omega_0 t)$$



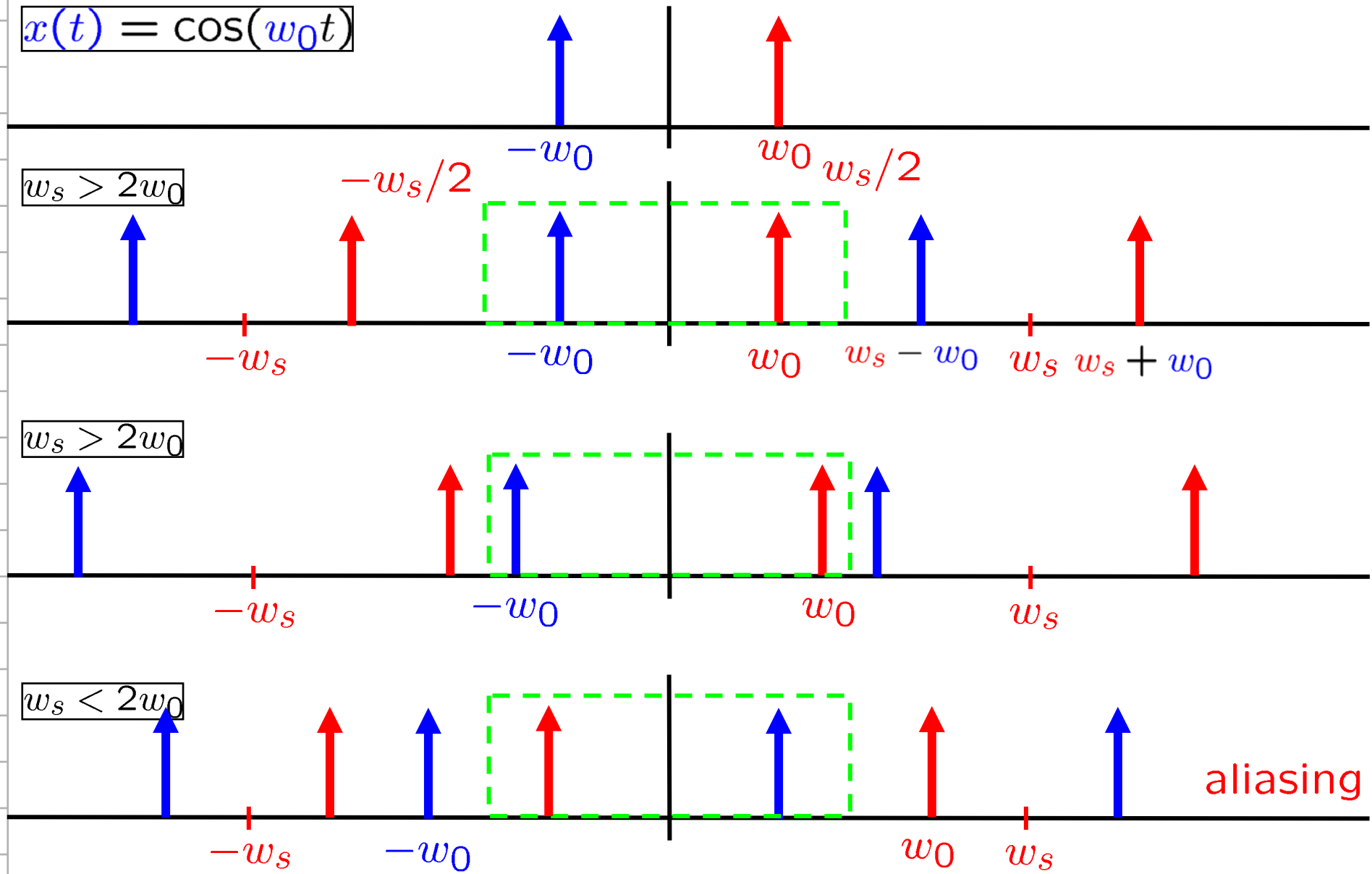
Overlapping in Frequency-Domain: Aliasing

$$x(t) = \cos(w_0 t)$$



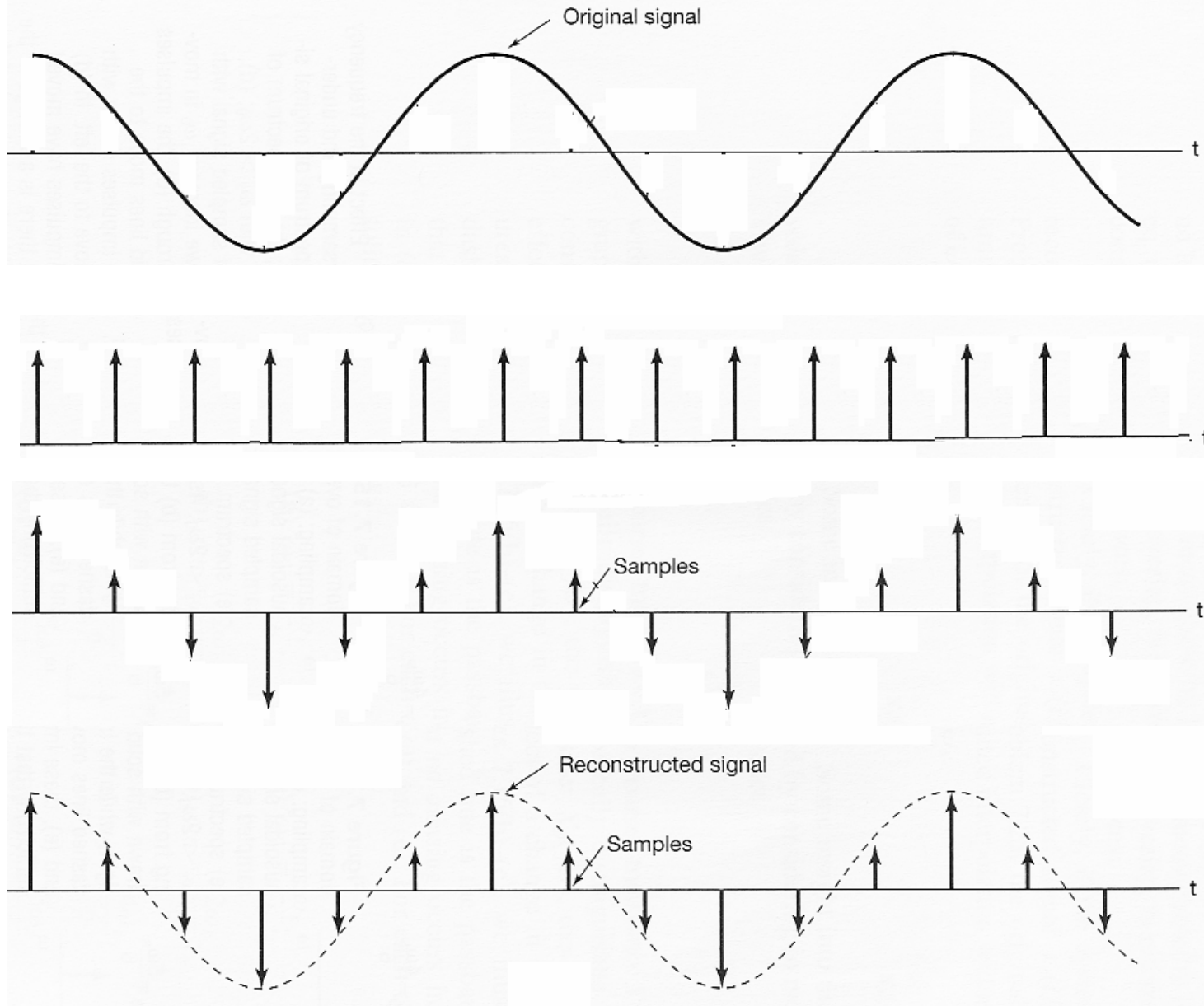
Overlapping in Frequency-Domain: Aliasing

$$x(t) = \cos(w_0 t)$$

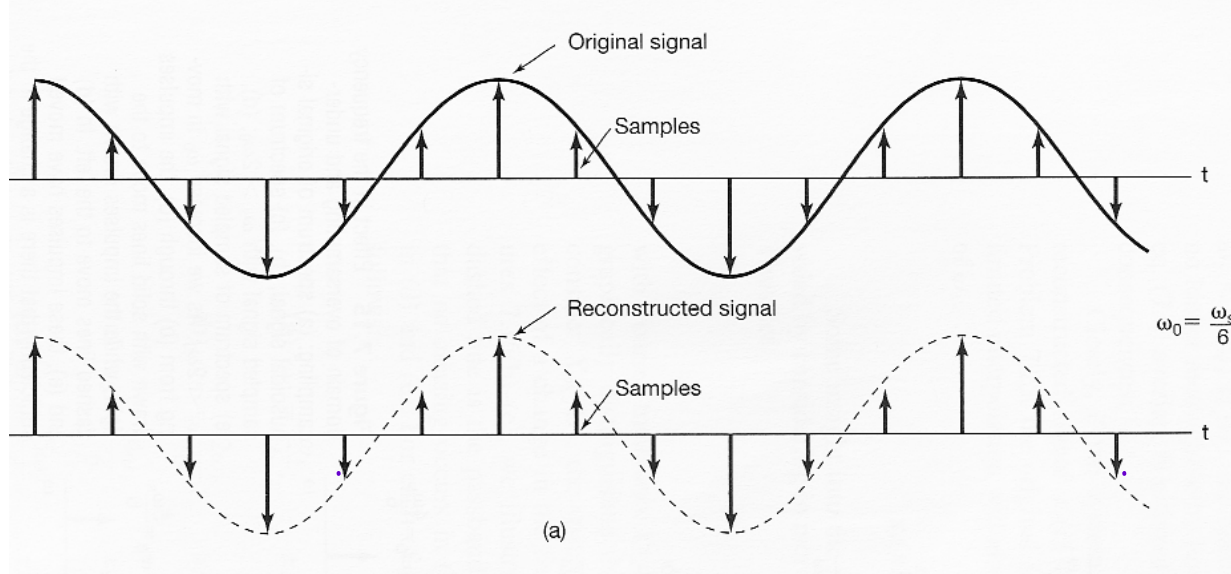


■ Overlapping in Frequency-Domain: Aliasing

$$\omega_0 = \frac{\omega_s}{6}$$



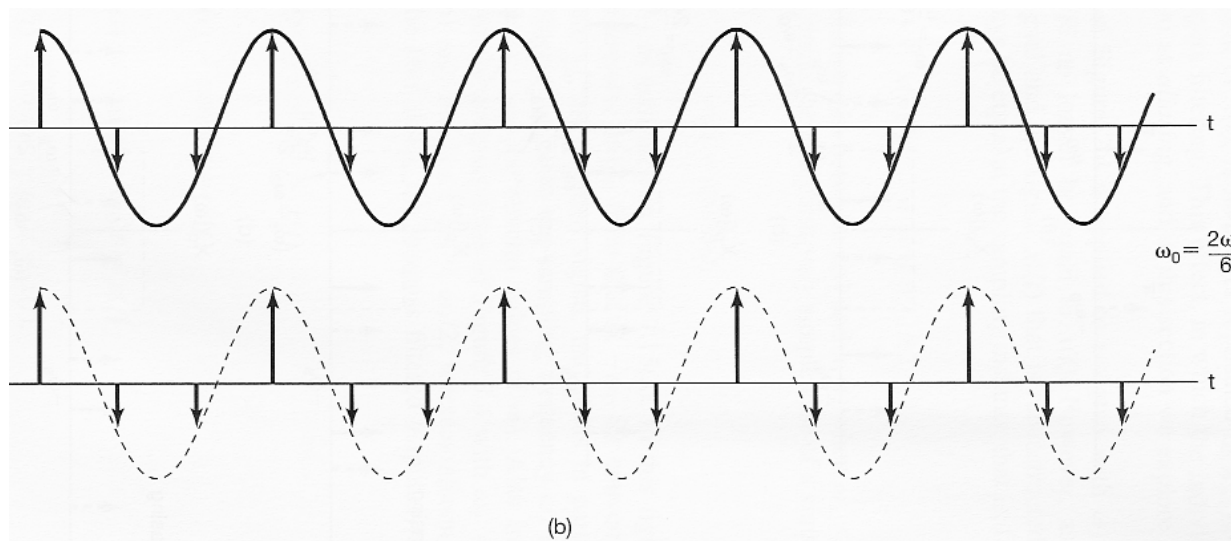
Overlapping in Frequency-Domain: Aliasing



signal frequency change
sampling freq. fix

$$\omega_0 = \frac{\omega_s}{6}$$

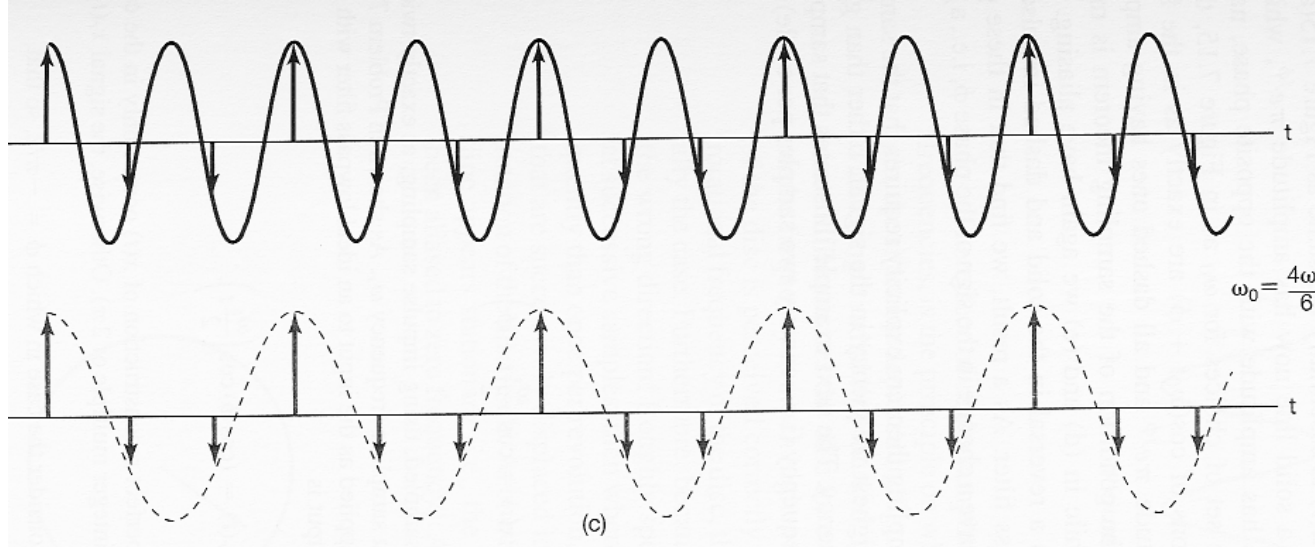
$$\omega_s = 6 \omega_0$$



$$\omega_0 = \frac{2\omega_s}{6}$$

$$\omega_s = 3 \omega_0 \quad \omega_s > 2\omega_0$$

Overlapping in Frequency-Domain: Aliasing

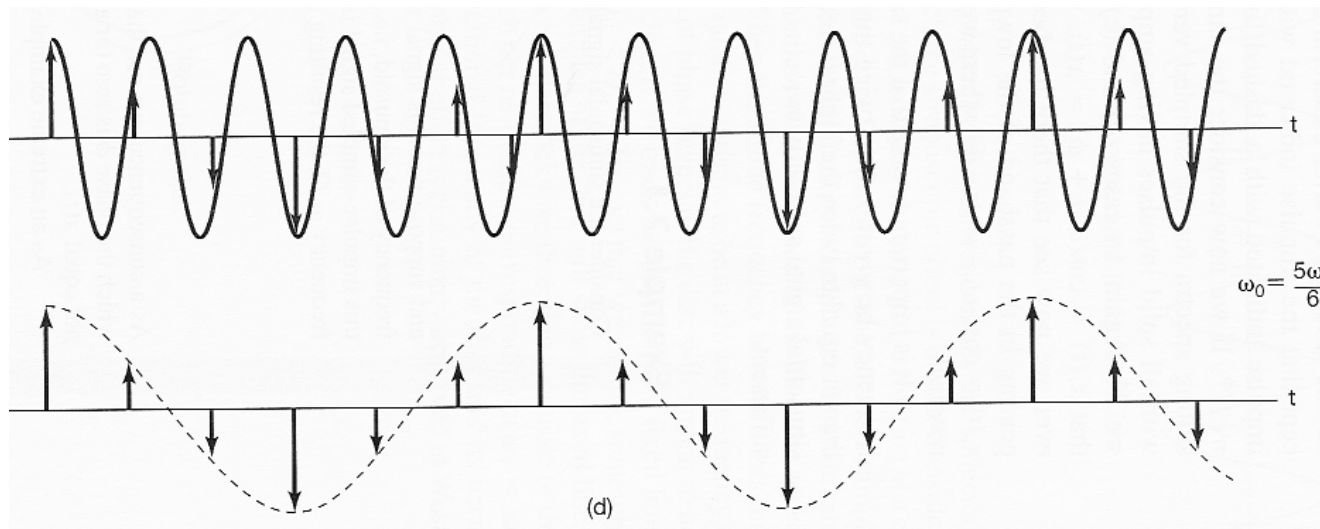


$$\omega_0 = \frac{4\omega_s}{6}$$

$$\omega_s = \frac{3}{2}\omega_0$$

$$\omega_s < 2\omega_0$$

Aliasing



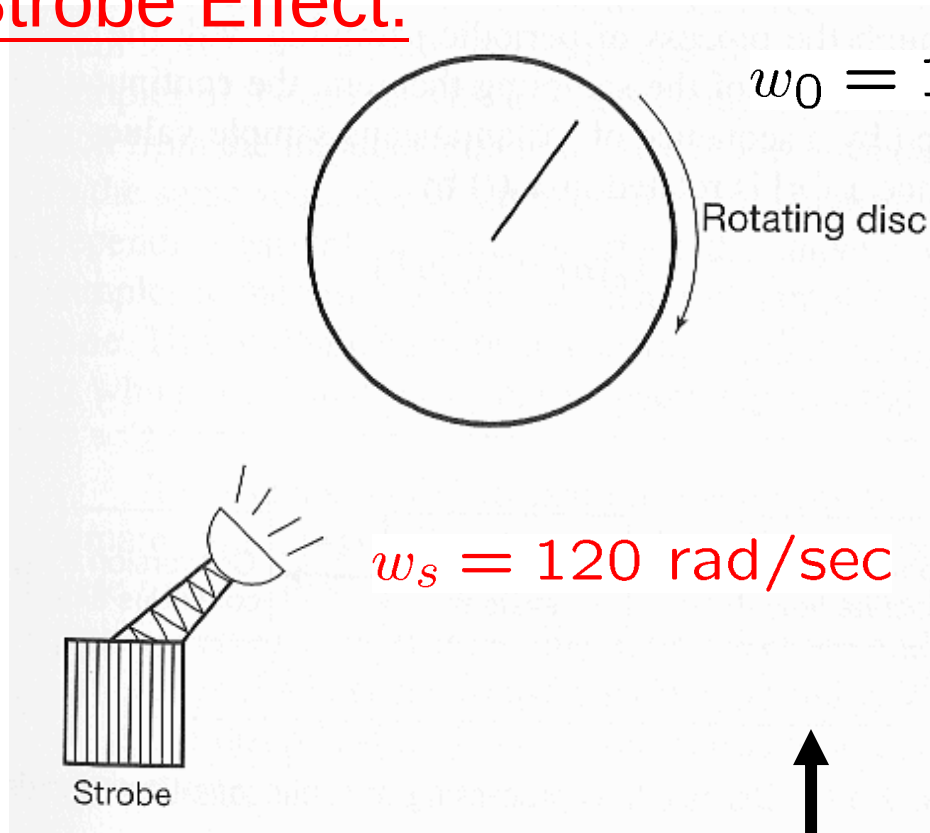
$$\omega_0 = \frac{5\omega_s}{6}$$

$$\omega_s = \frac{6}{5}\omega_0$$

$$\omega_s < 2\omega_0$$

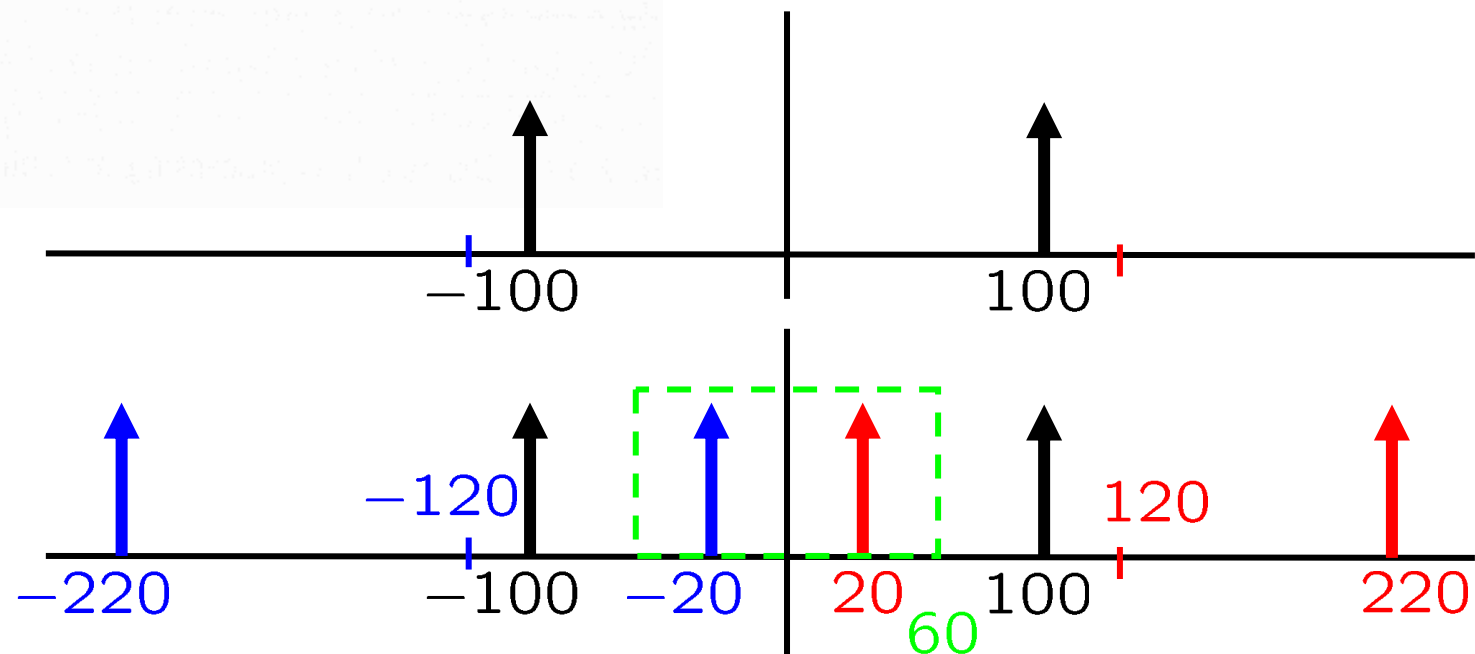
Aliasing

■ Strobe Effect:



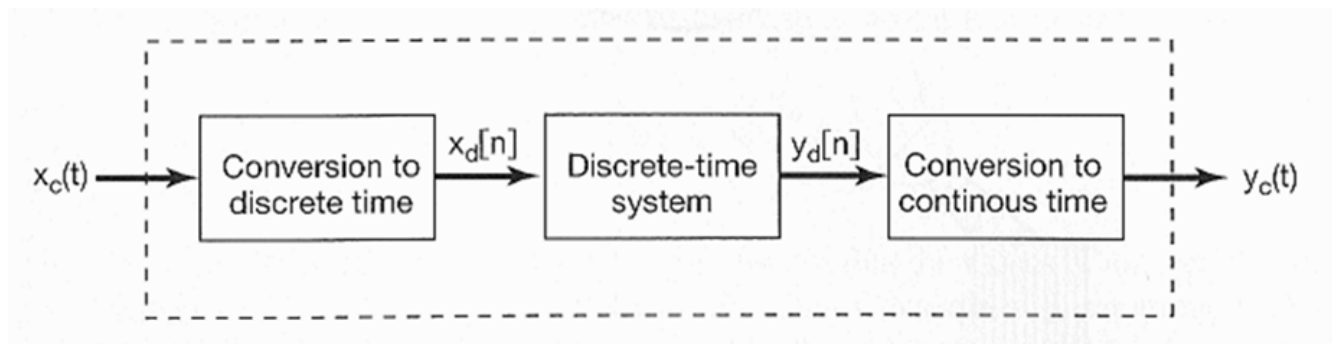
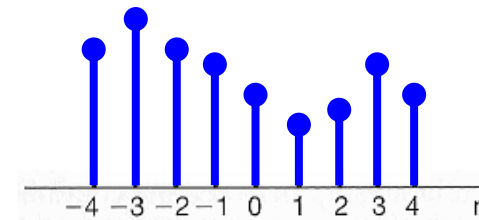
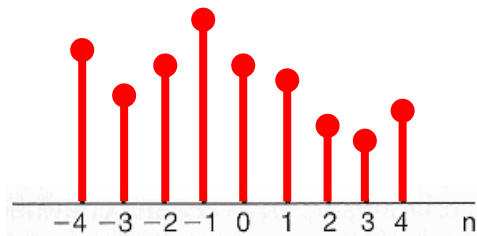
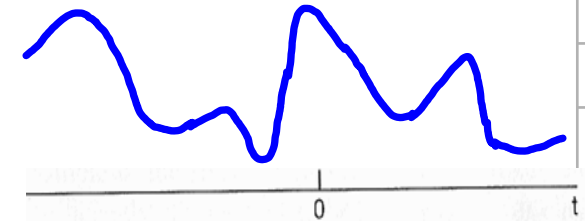
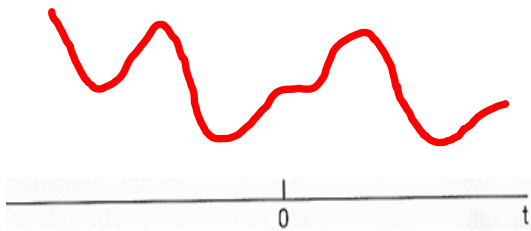
$$\Rightarrow w = \pm w_s \pm w_0$$

$$= +20, -20$$

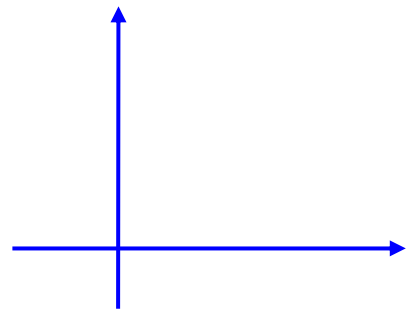
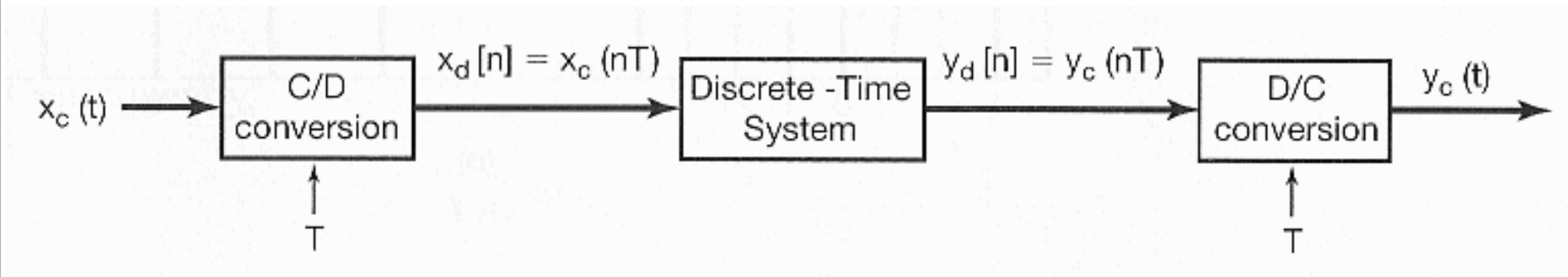


- Representation of of a Continuous-Time Signal by Its Samples: The Sampling Theorem
- Reconstruction of of a Signal from Its Samples Using Interpolation
- The Effect of Under-sampling: Aliasing
- Discrete-Time Processing of Continuous-Time Signals
- Sampling of Discrete-Time Signals

■ Discrete-Time Processing of CT Signals:

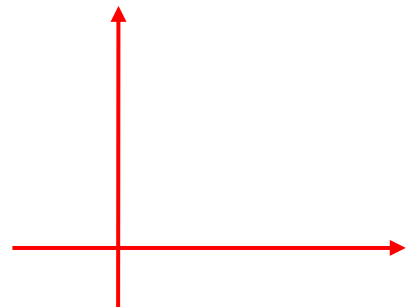


■ C/D or A-to-D (ADC) and D/C or D-to-A (DAC):



C/D: continuous-to-discrete-time conversion

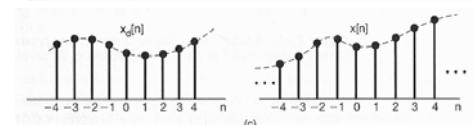
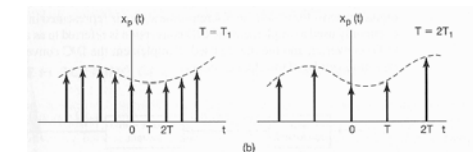
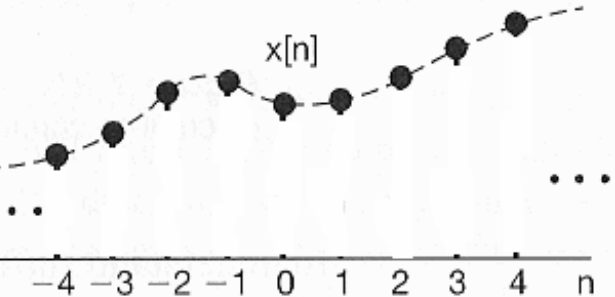
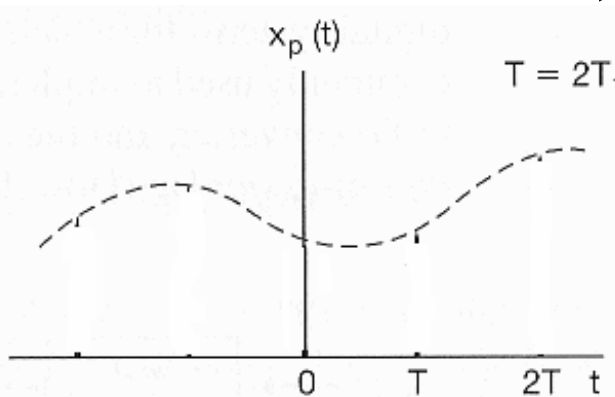
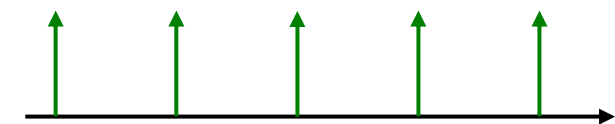
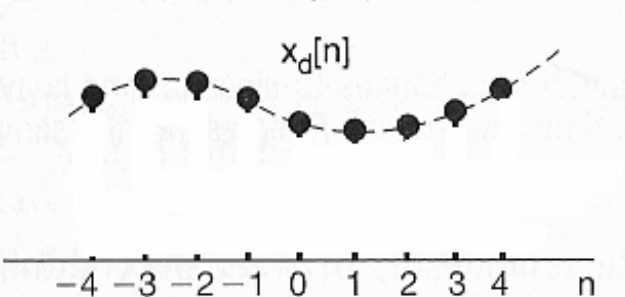
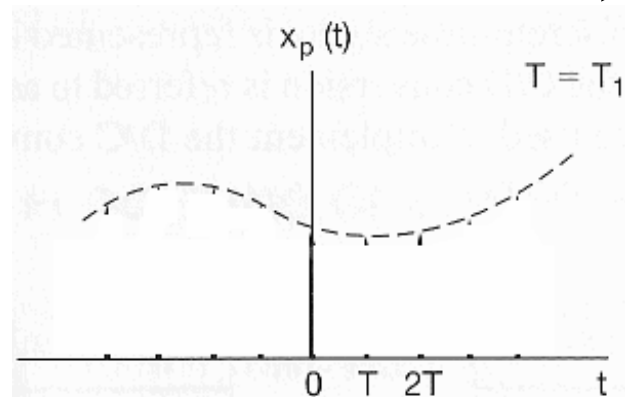
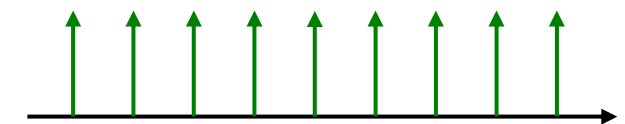
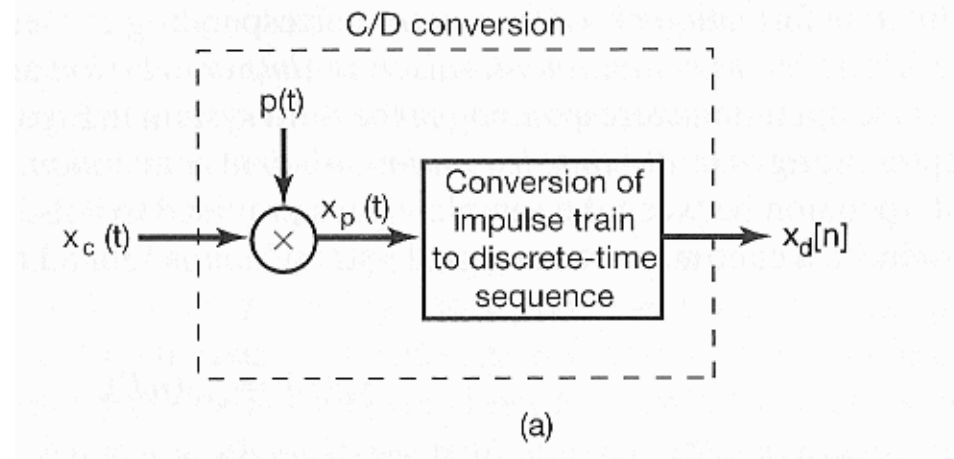
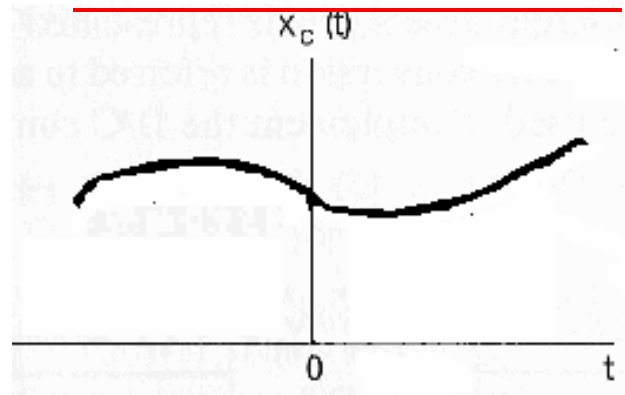
A-to-D: analog-to-digital converter



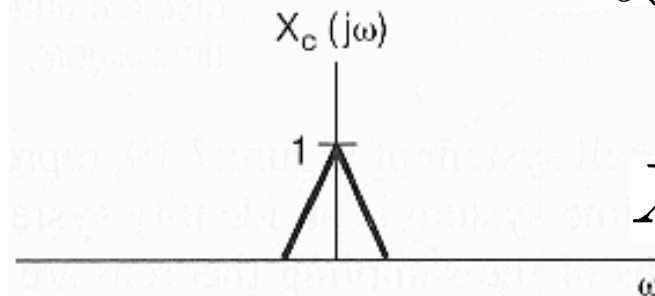
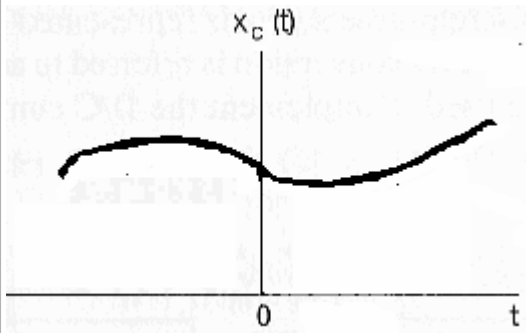
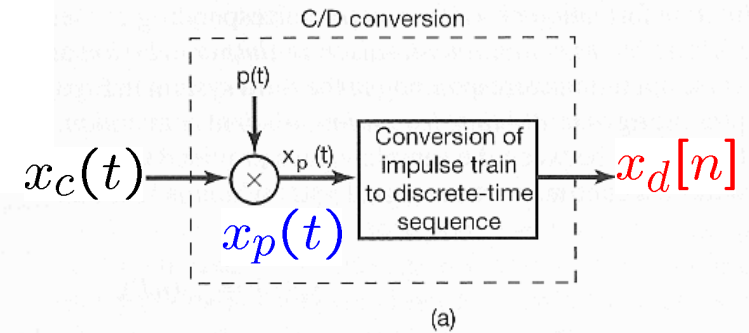
D/C: discrete-to-continuous-time conversion

D-to-A: digital-to-analog converter

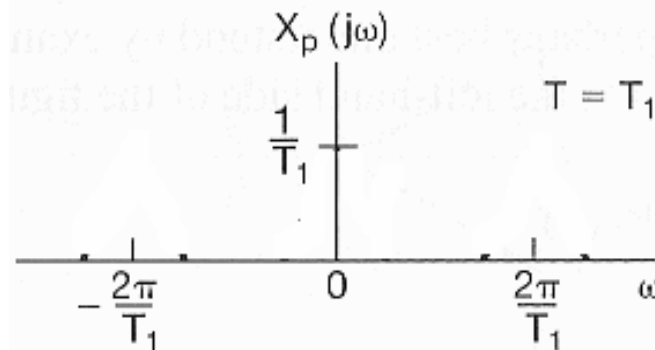
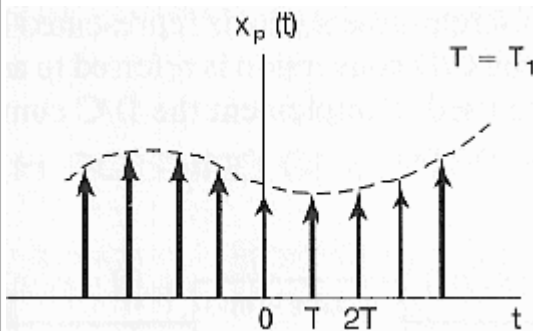
C/D Conversion:



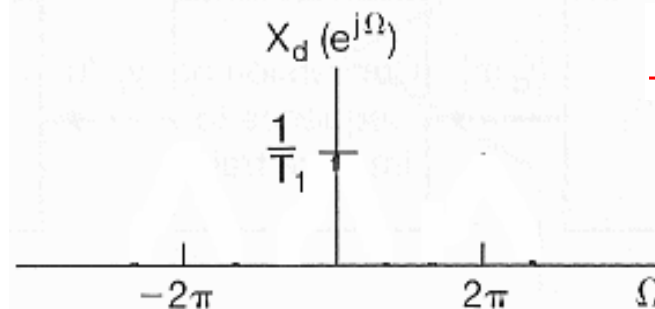
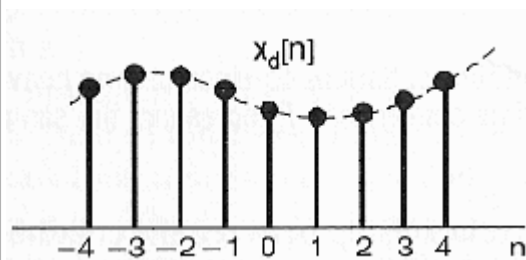
■ C/D Conversion:



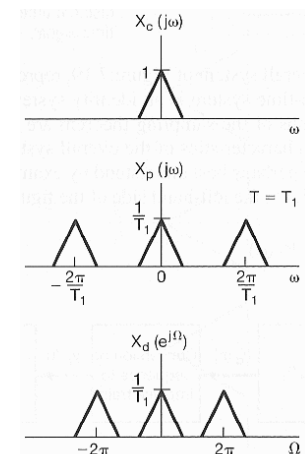
$$X_c(j\omega)$$



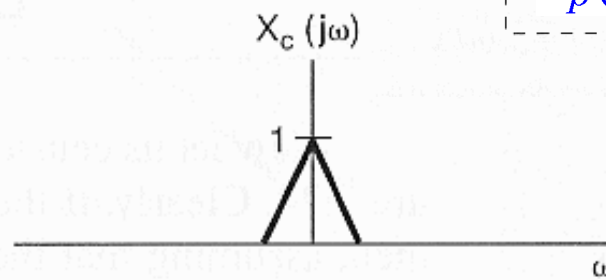
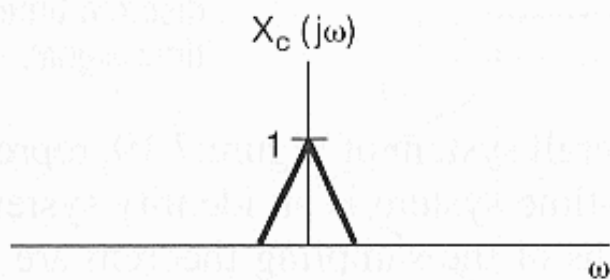
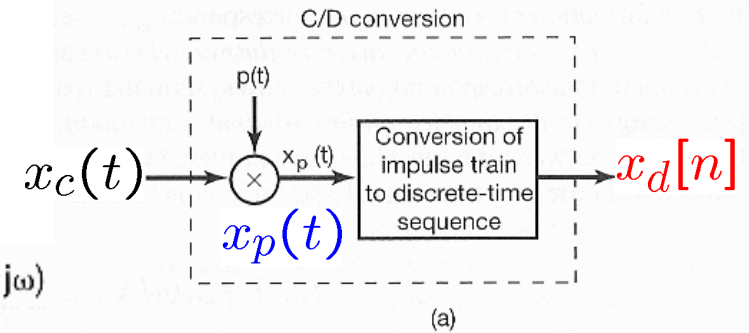
$$X_p(j\omega)$$



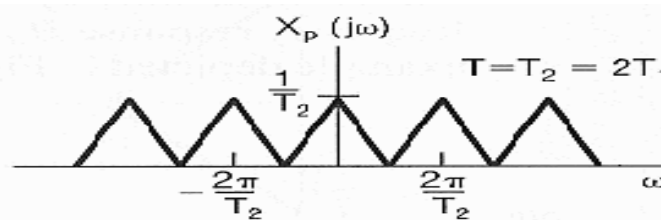
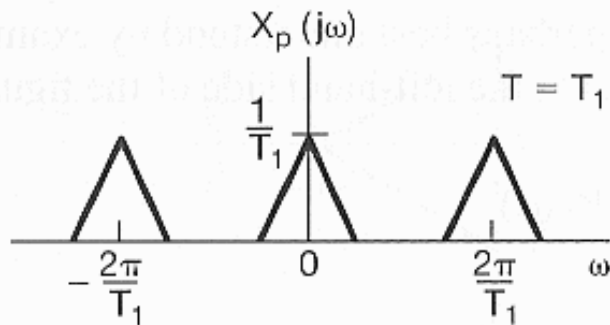
$$X_d(e^{j\Omega})$$



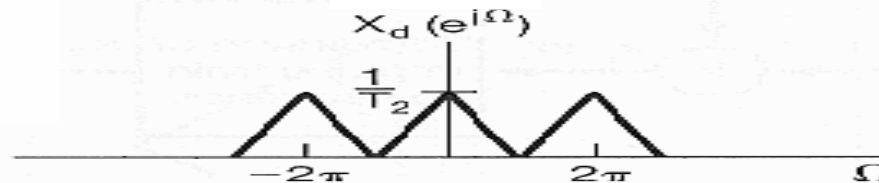
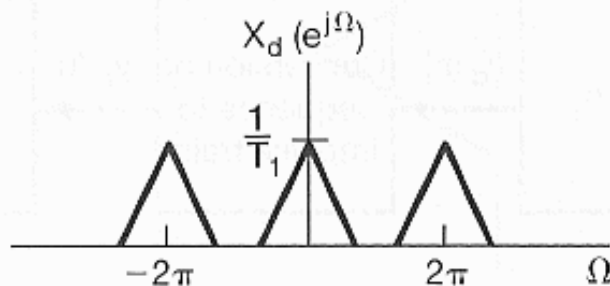
■ C/D Conversion:



$$X_c(j\omega)$$



$$X_p(j\omega)$$



$$X_d(e^{j\Omega})$$

■ C/D Conversion:

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT)$$

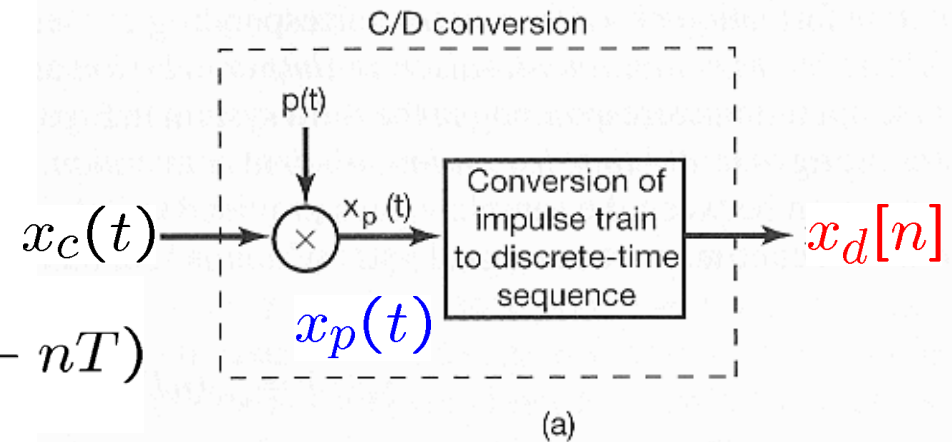


Table 4.2, p. 329

$$\delta(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0}$$

Eq 7.3, 7.6, p. 517

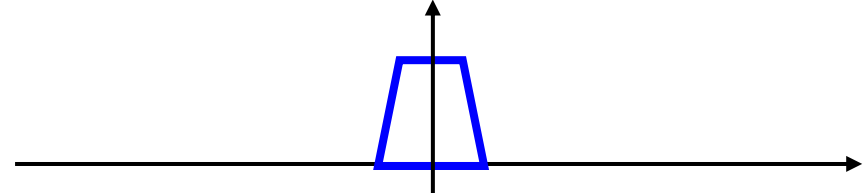
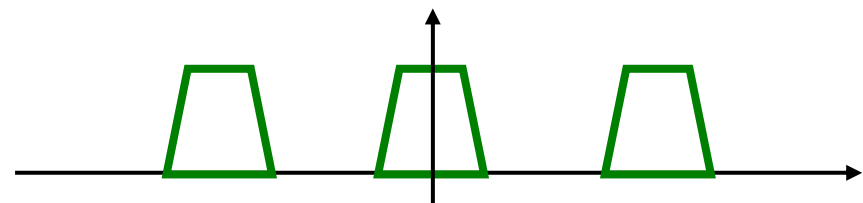
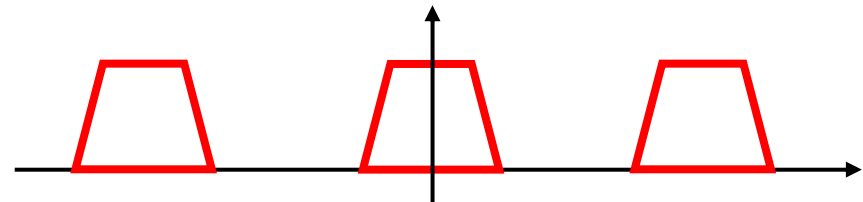
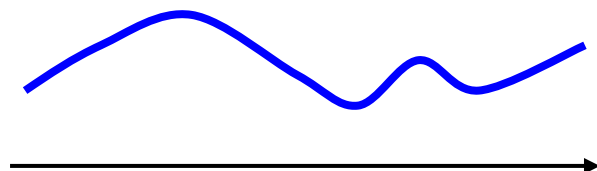
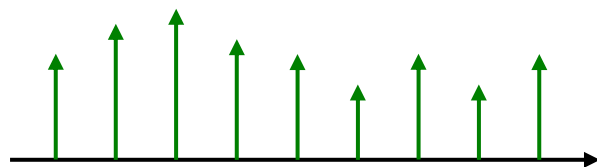
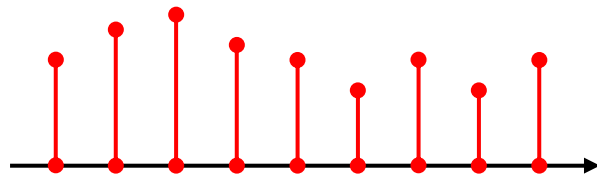
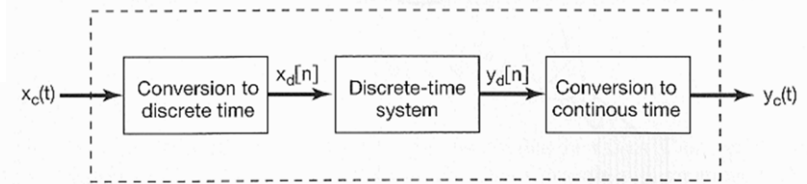
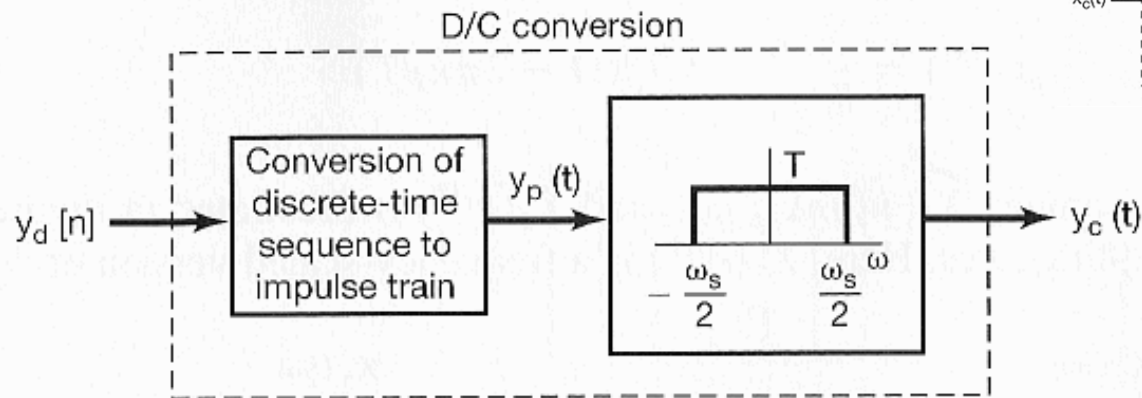
$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\omega nT} = \frac{1}{T} \sum_{K=-\infty}^{+\infty} X_c(j(\omega - K\omega_s))$$

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n} = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n}$$

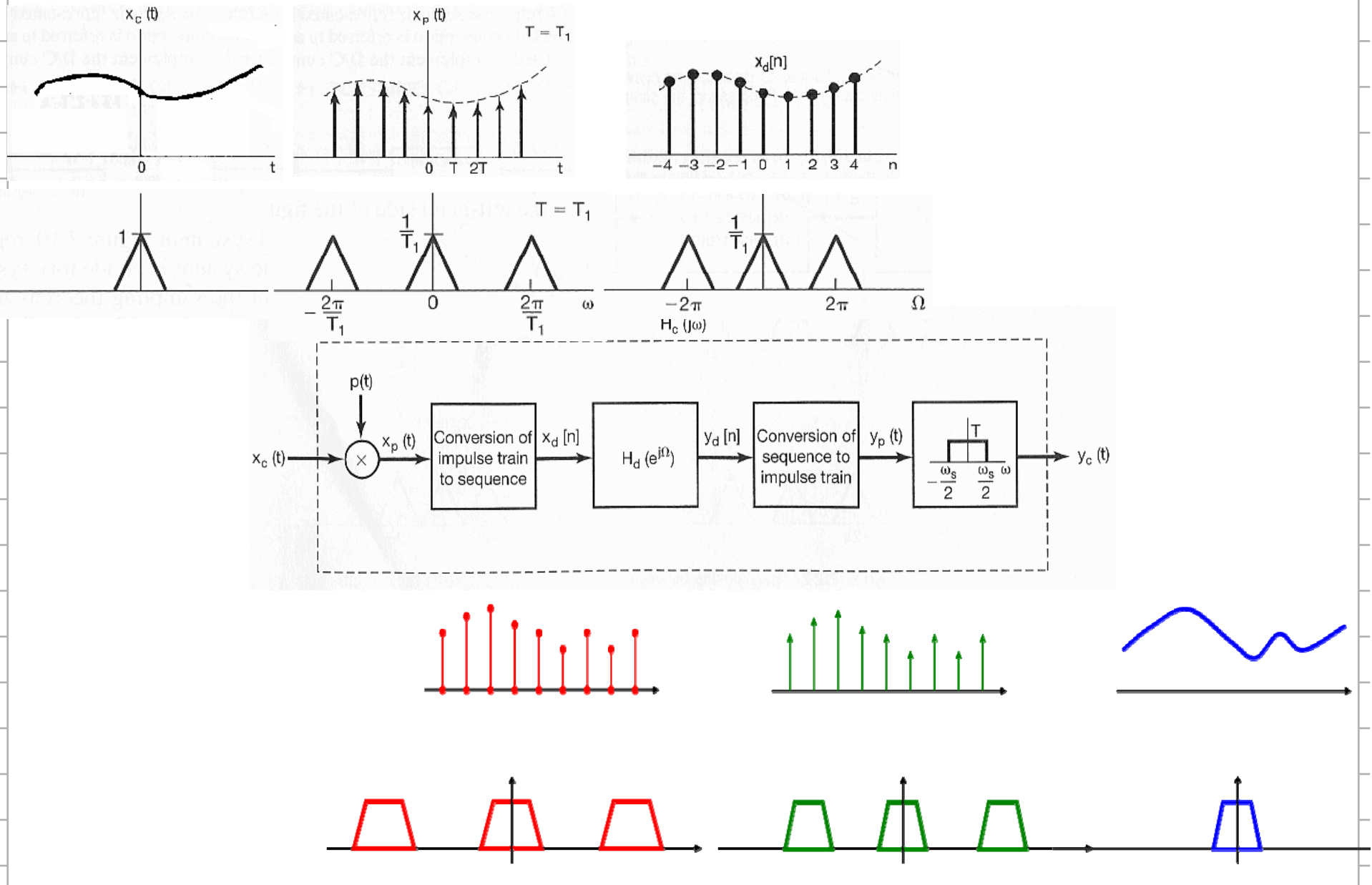
$$\Rightarrow X_d(e^{j\Omega}) = X_p\left(j\frac{\Omega}{T}\right)$$

$$= \frac{1}{T} \sum_{K=-\infty}^{+\infty} X_c\left(j\left(\frac{\Omega}{T} - K\frac{2\pi}{T}\right)\right)$$

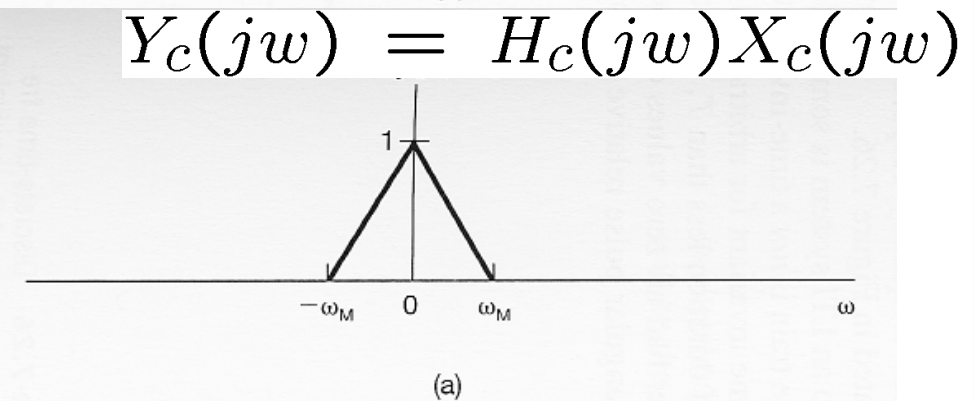
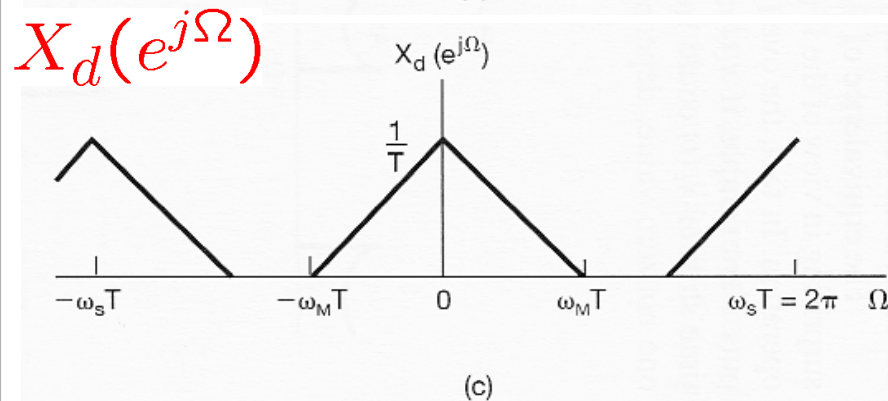
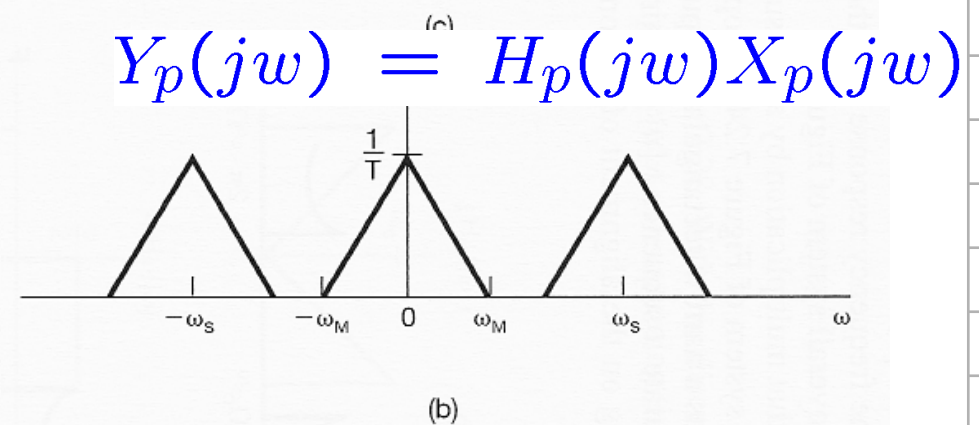
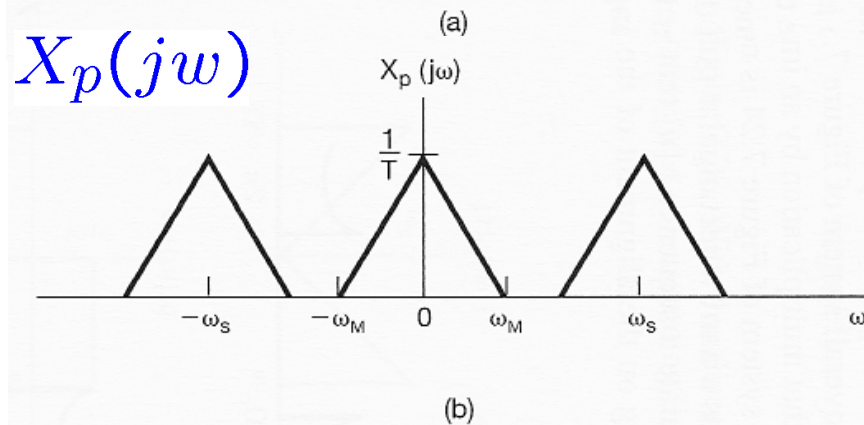
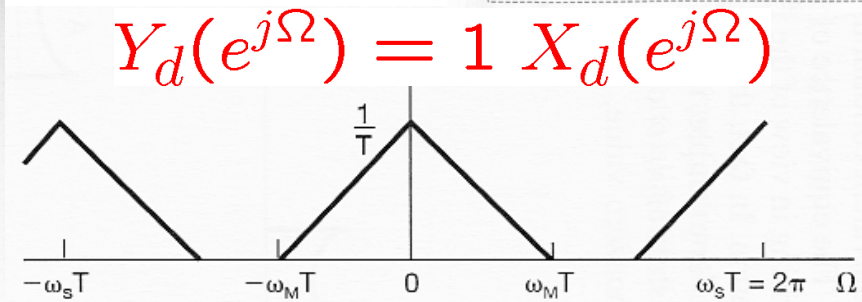
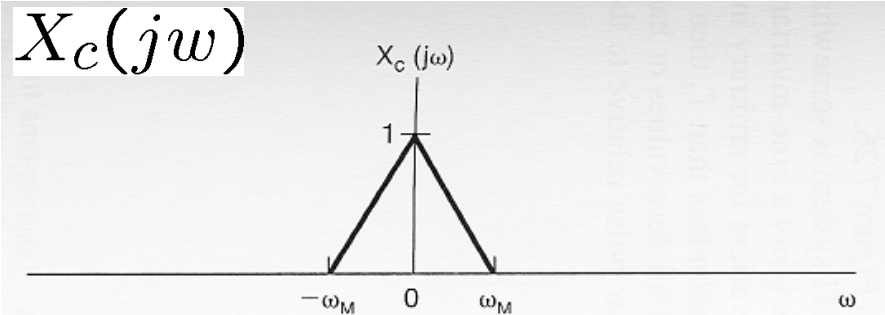
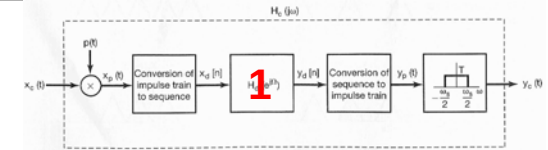
D/C Conversion:



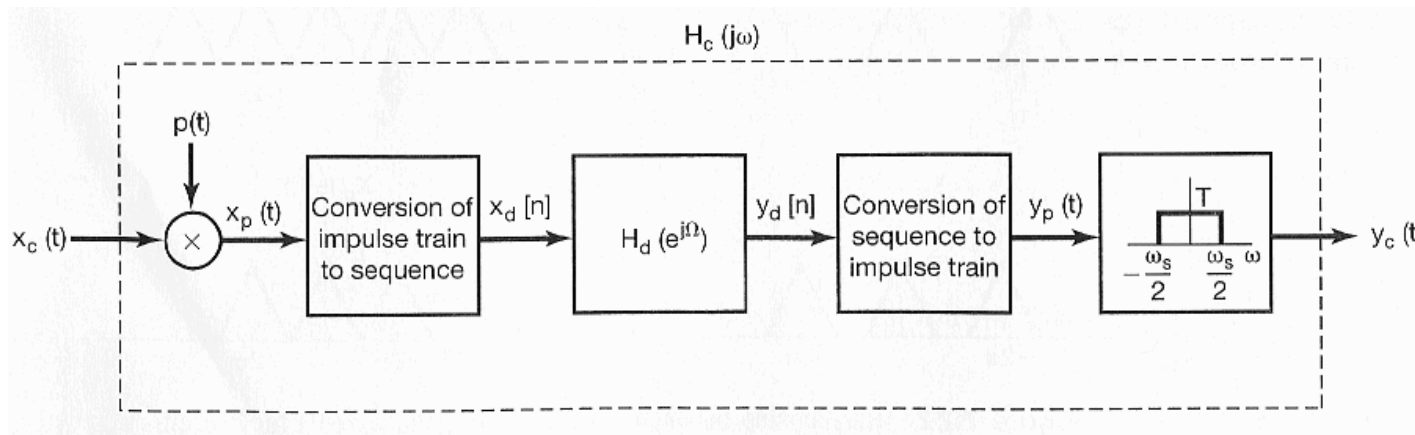
Overall System:



Frequency-Domain Illustration:

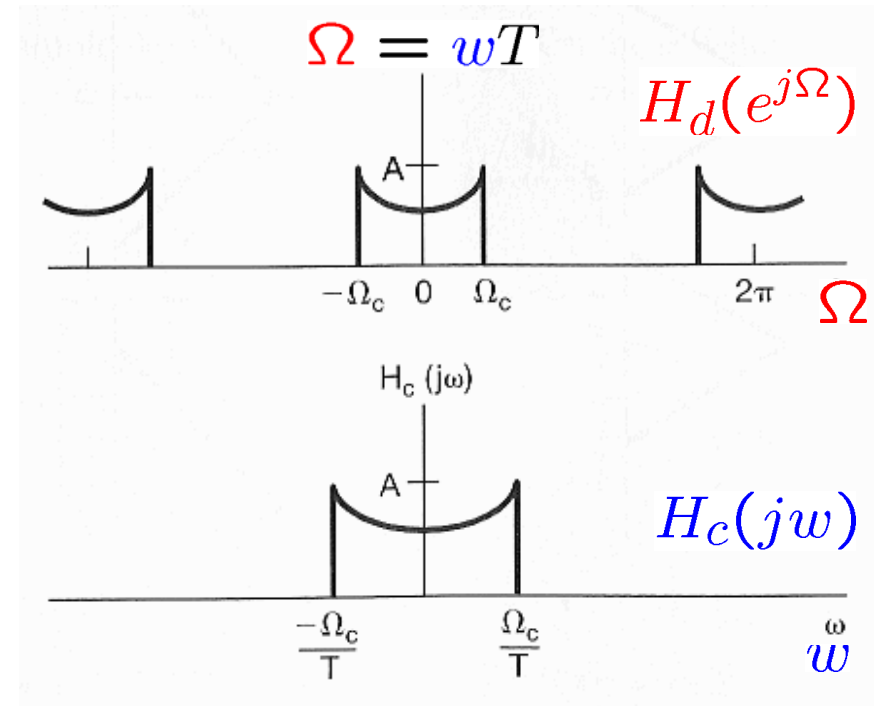


CT & DT Frequency Responses:

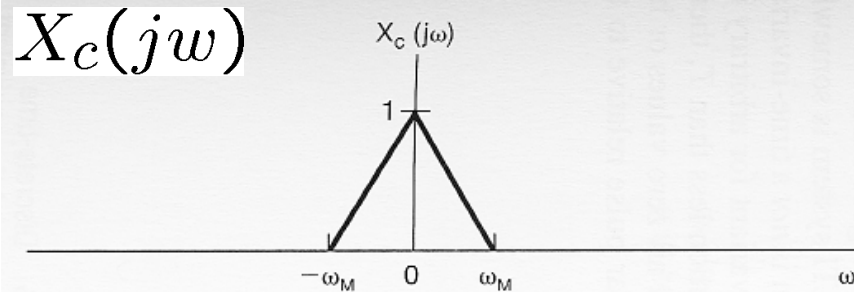
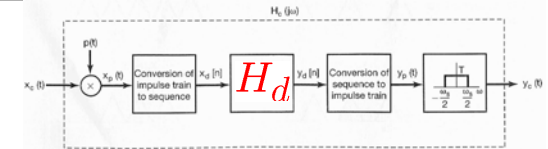


$$Y_c(jw) = X_c(jw)H_c(jw)$$

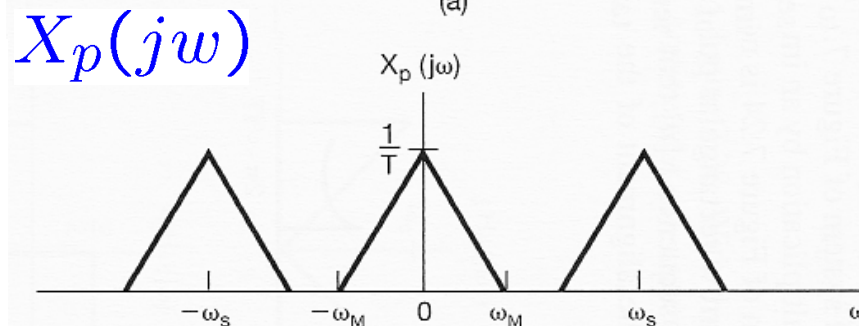
$$H_c(jw) = \begin{cases} H_d(e^{jwT}), & |w| < w_s/2 \\ 0, & |w| > w_s/2 \end{cases}$$



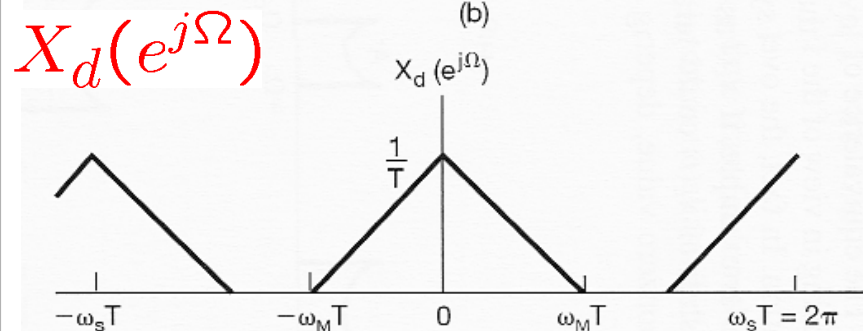
Frequency-Domain Illustration:



(a)

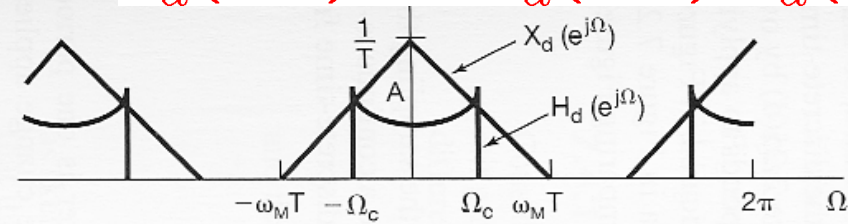


(b)



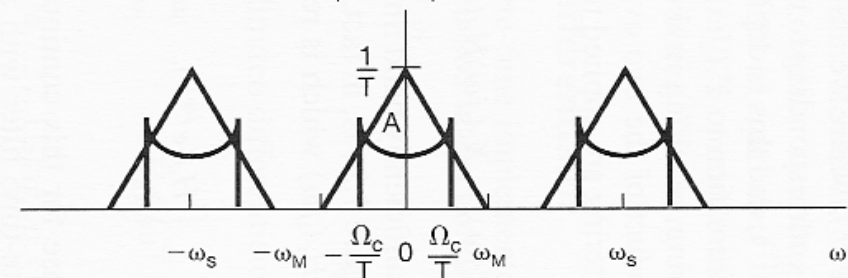
(c)

$$Y_d(e^{j\Omega}) = H_d(e^{j\Omega}) X_d(e^{j\Omega})$$

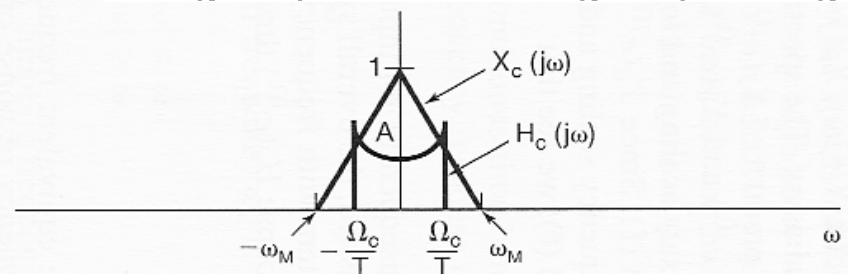


(d)

$$Y_p(j\omega) = H_p(j\omega) X_p(j\omega)$$



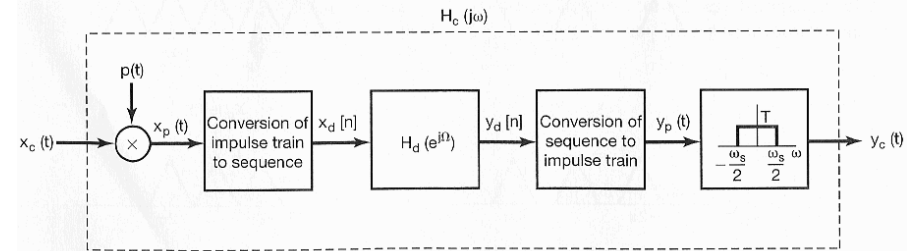
$$Y_c(j\omega) = H_c(j\omega) X_c(j\omega)$$



(f)

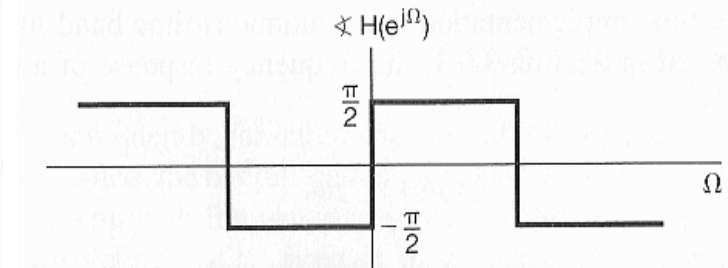
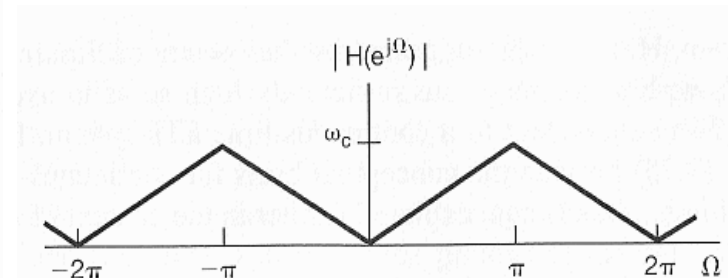
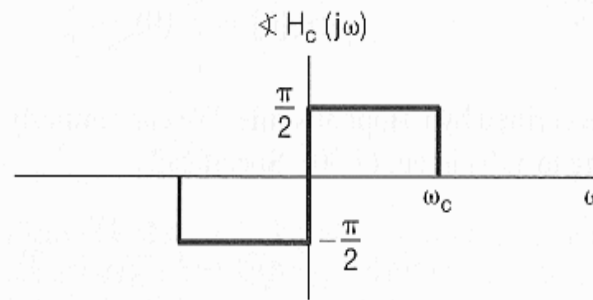
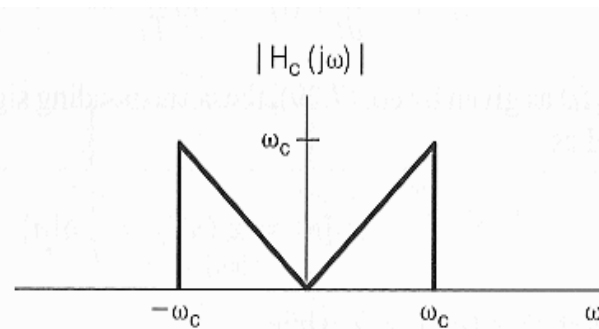
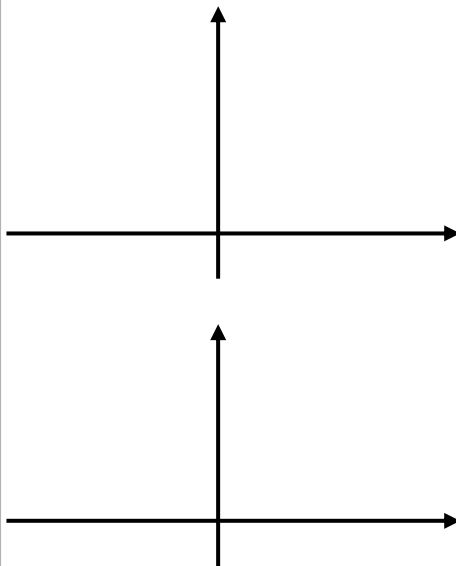
■ Digital Differentiator:

Ex 4.16, p. 317



$$H_c(j\omega) = \begin{cases} j\omega, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

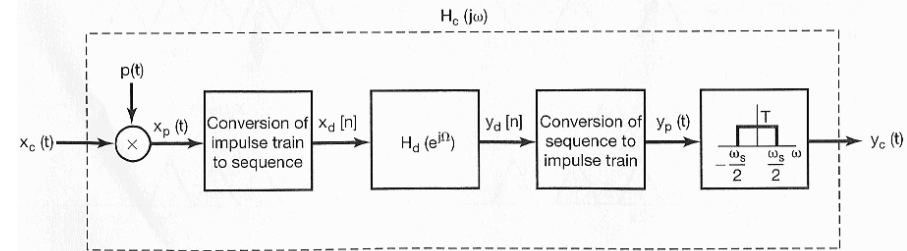
$$H_d(e^{j\Omega}) = j\left(\frac{\Omega}{T}\right), \quad |\Omega| < \pi$$



$$\Omega = \omega T, \quad \omega_s = 2\omega_c$$

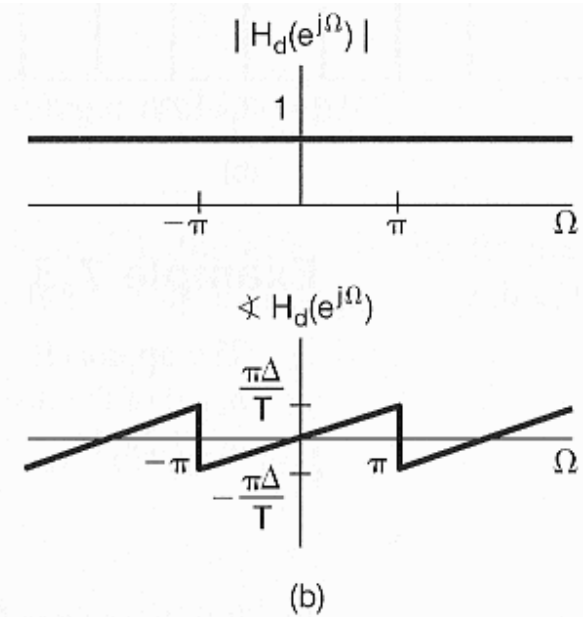
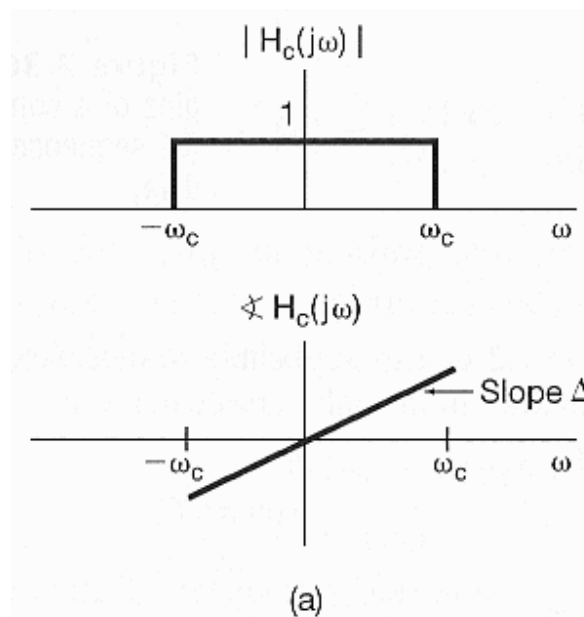
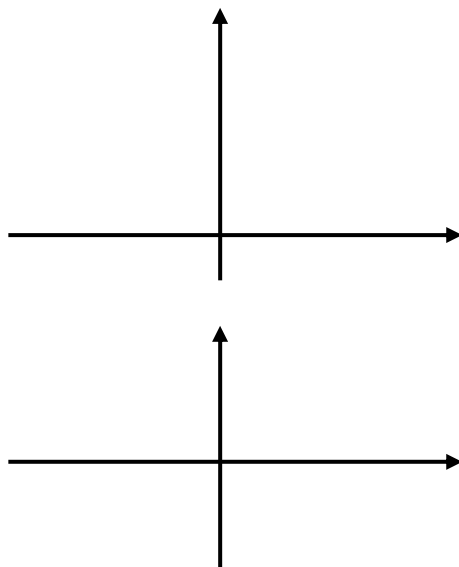
■ Half-Sample Delay:

Ex 4.15, p. 317



$$H_c(j\omega) = \begin{cases} e^{-j\omega\Delta}, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

$$H_d(e^{j\Omega}) = e^{-j\Omega\Delta/T}, \quad |\Omega| < \pi$$



$$\Omega = \omega T, \quad \omega_s = 2\omega_c$$

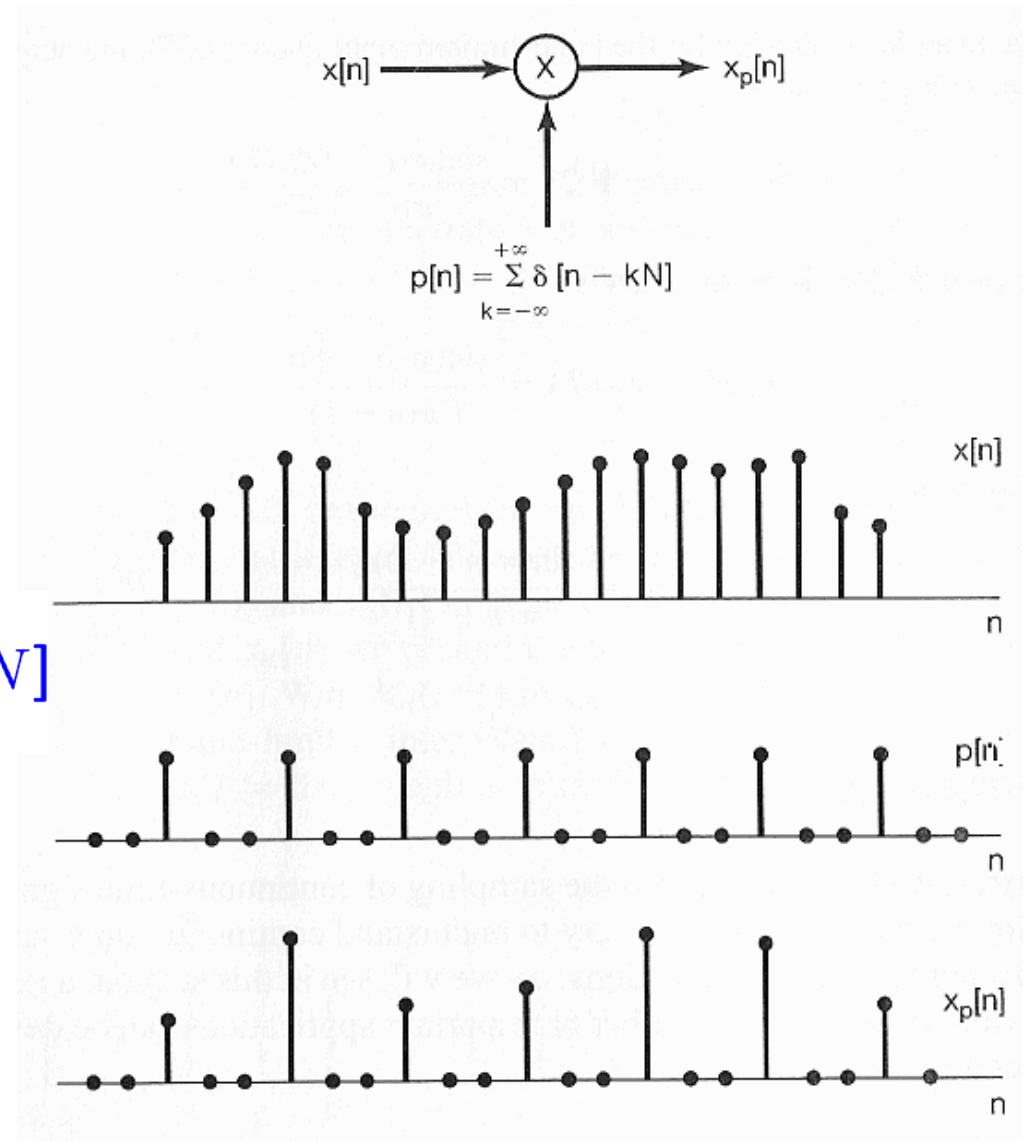
- Representation of of a Continuous-Time Signal by Its Samples: The Sampling Theorem
- Reconstruction of of a Signal from Its Samples Using Interpolation
- The Effect of Undersampling: Aliasing
- Discrete-Time Processing of Continuous-Time Signals
- Sampling of Discrete-Time Signals

■ Impulse-Train Sampling:

$$x_p[n] = \begin{cases} x[n], & \text{if } n = kN \\ 0, & \text{otherwise} \end{cases}$$

$$x_p[n] = x[n] p[n]$$

$$= \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$

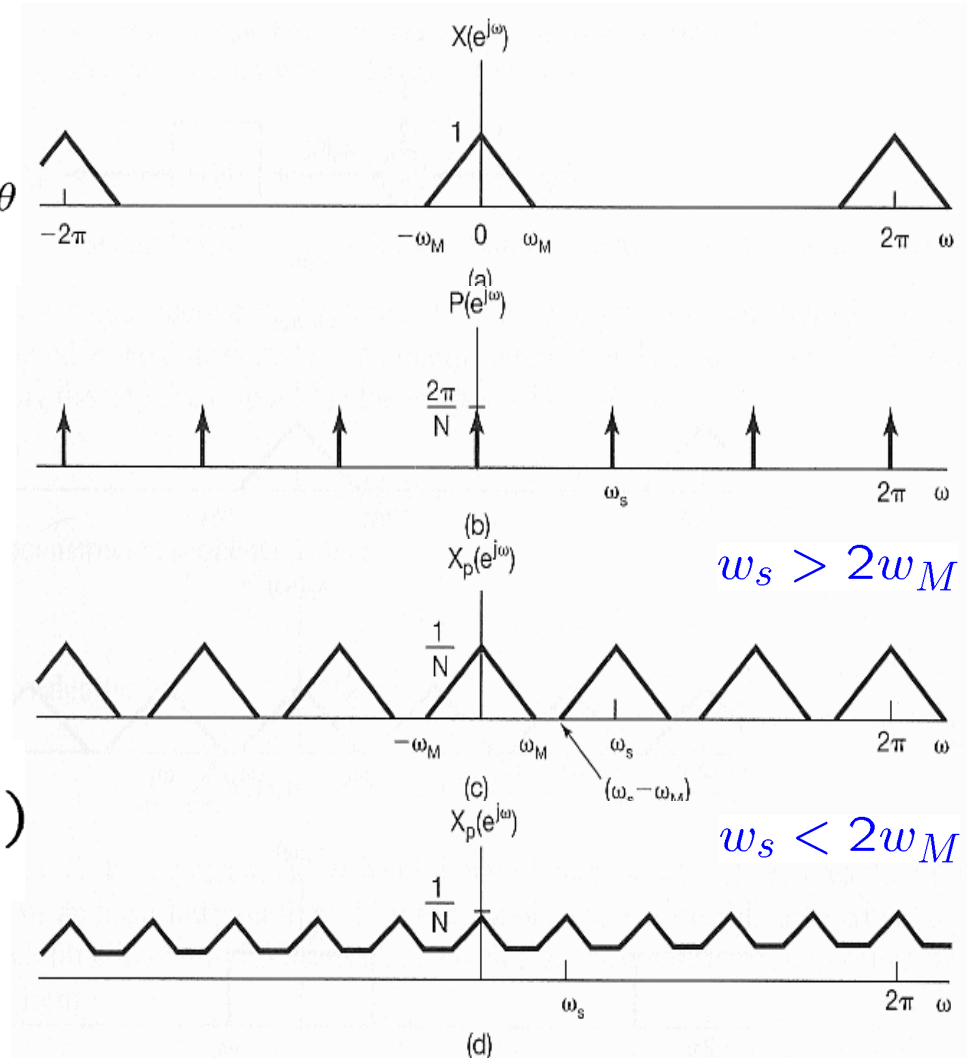


■ Impulse-Train Sampling:

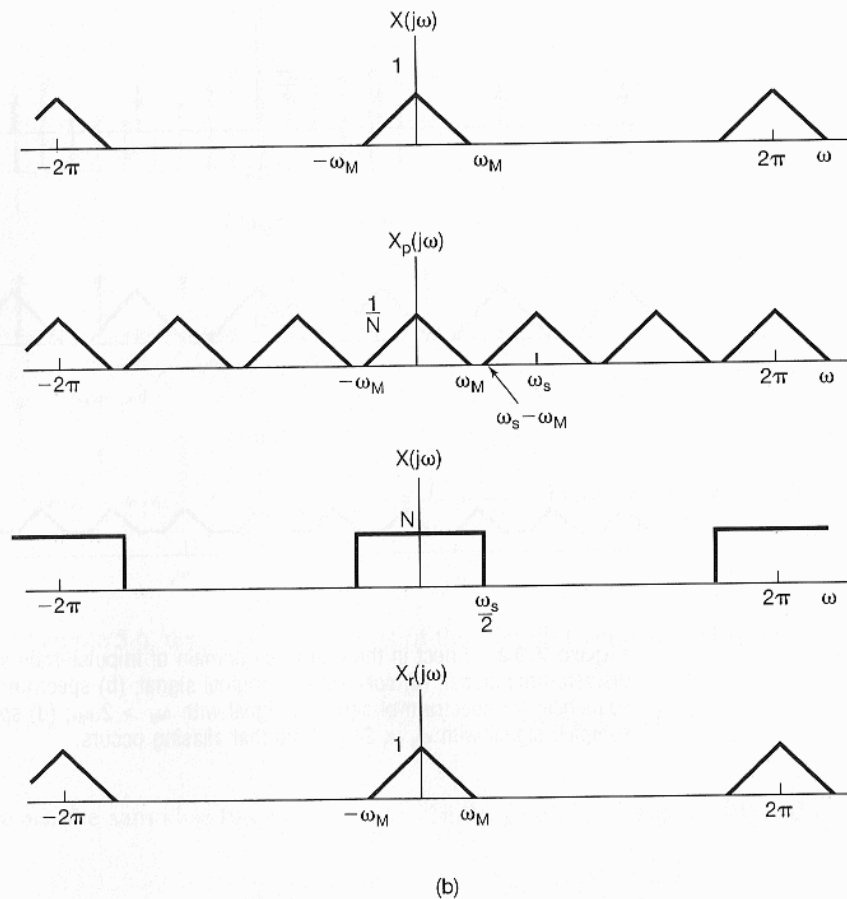
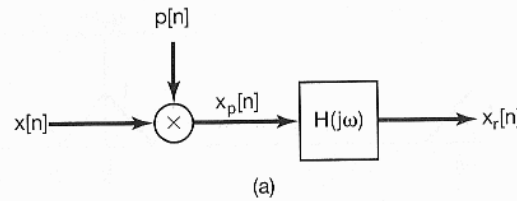
$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta$$

$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

$$\Rightarrow X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$

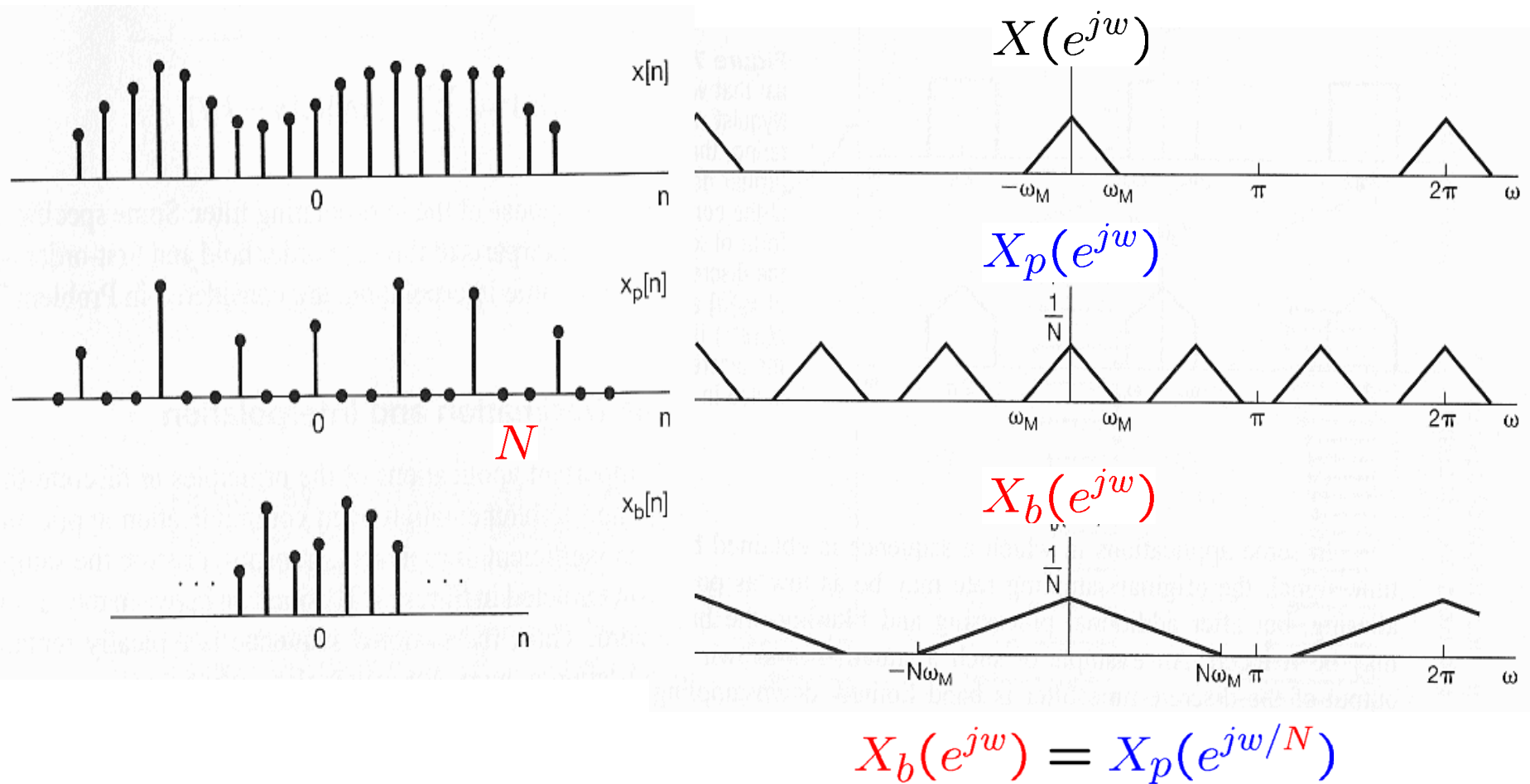


Exact Recovery Using Ideal Lowpass Filter:

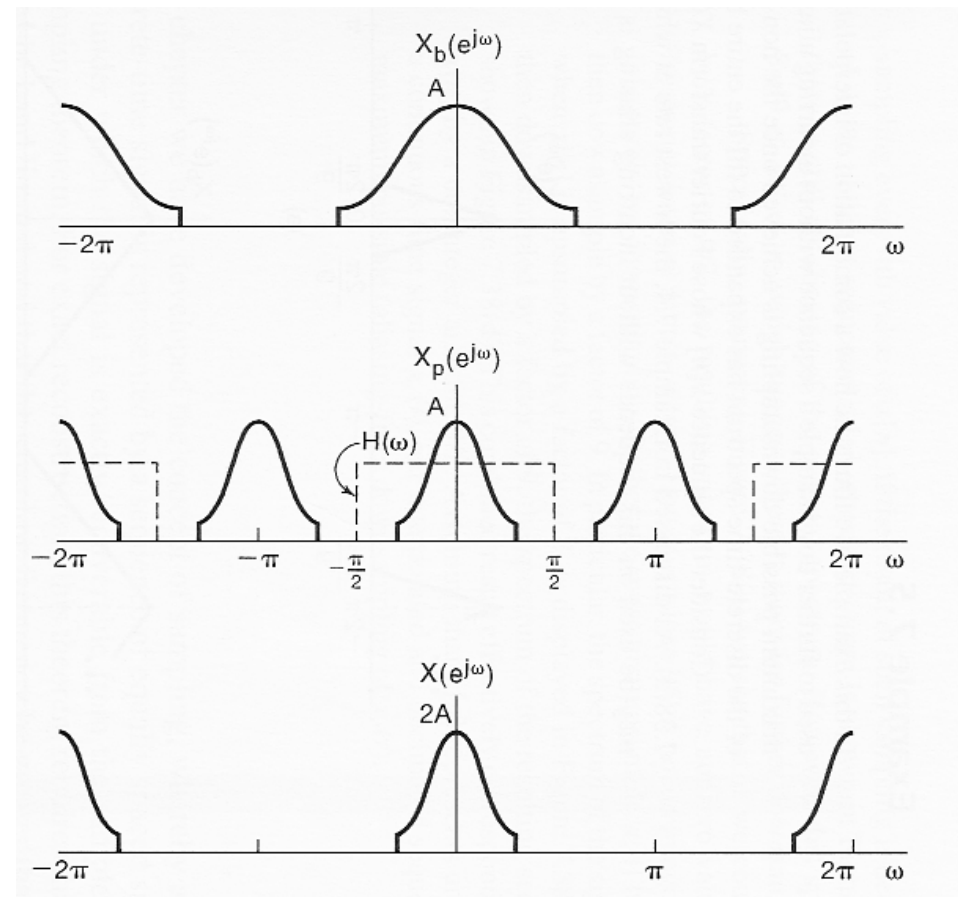
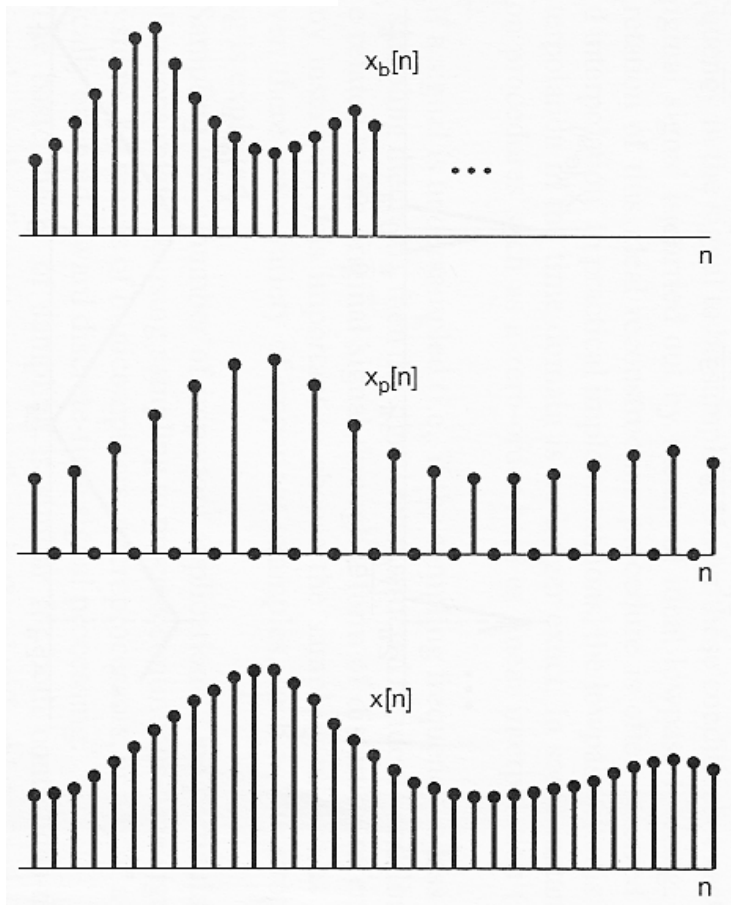
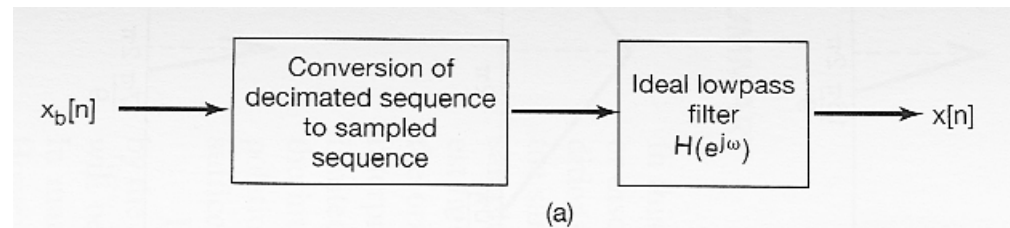


DT Decimation & Interpolation: Down-sampling

Eq 5.45, p. 378: Time expansion



Higher Equivalent Sampling Rate: Up-sampling



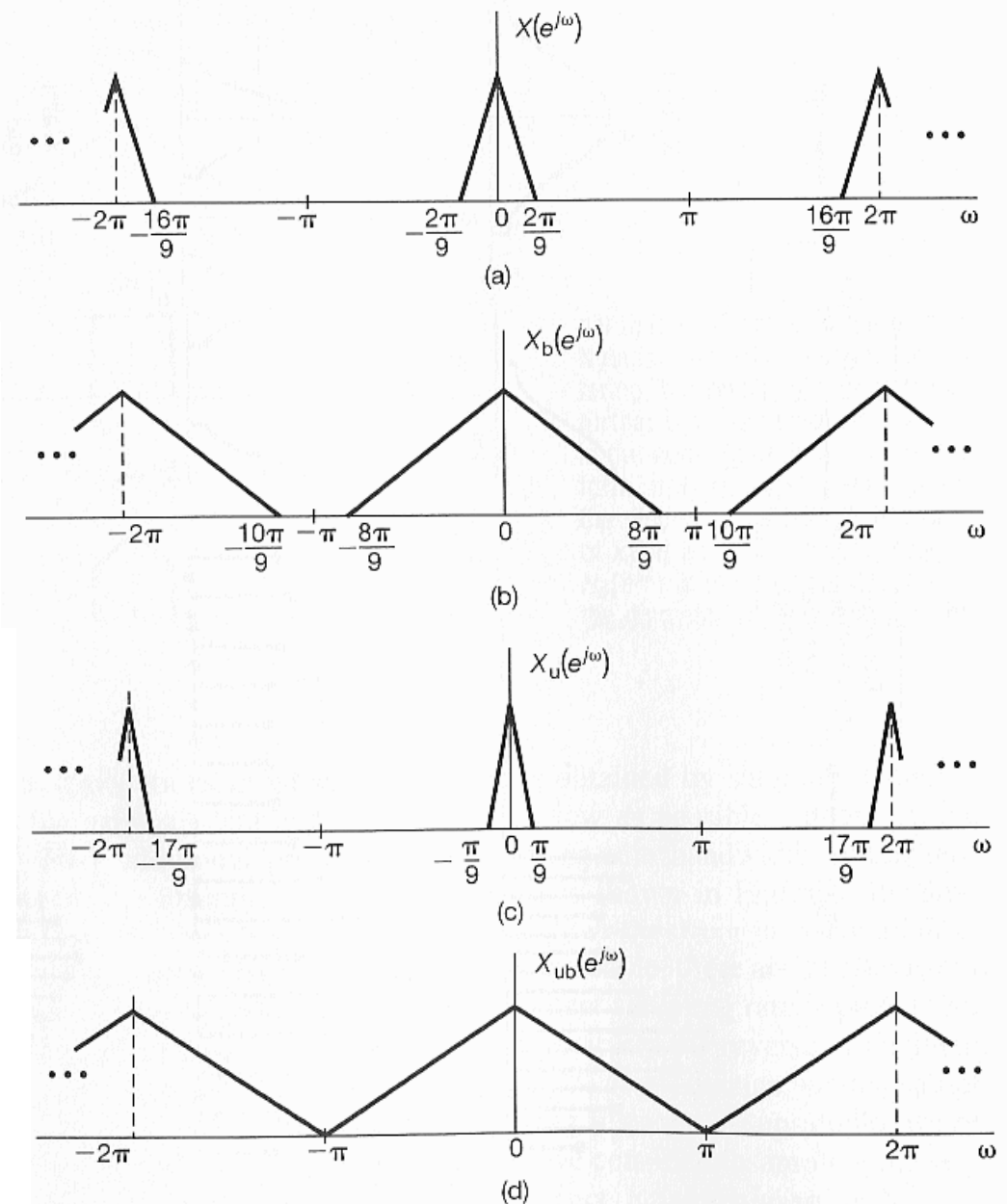
■ Down-sampling + Up-sampling:

$$\frac{2\pi}{9} \times 4 < \pi$$

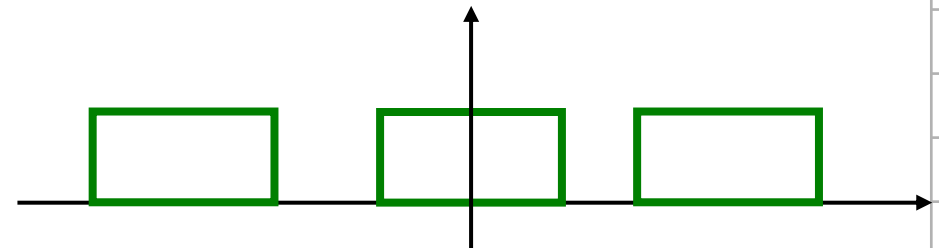
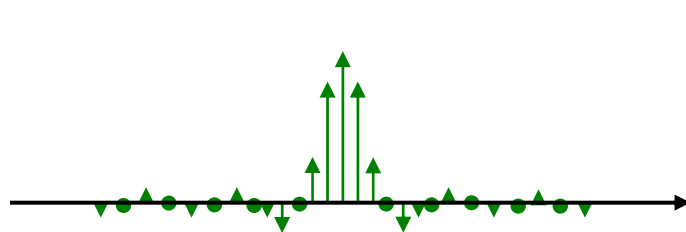
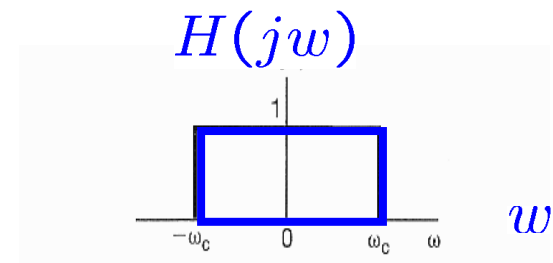
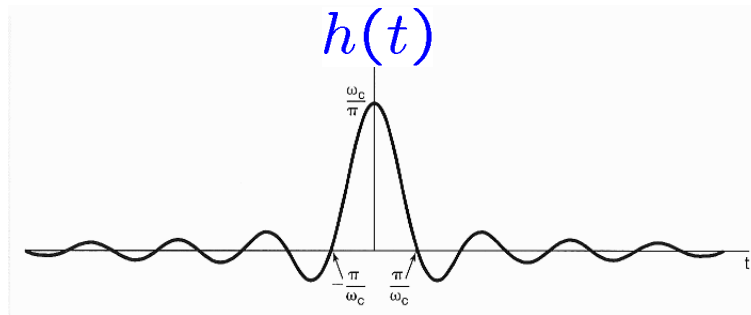
$$\frac{2\pi}{9} \times \frac{9}{2} = \pi$$

$$\frac{2\pi}{9} \times \frac{1}{2} = \frac{\pi}{9}$$

$$\frac{\pi}{9} \times 9 = \pi$$



In Summary

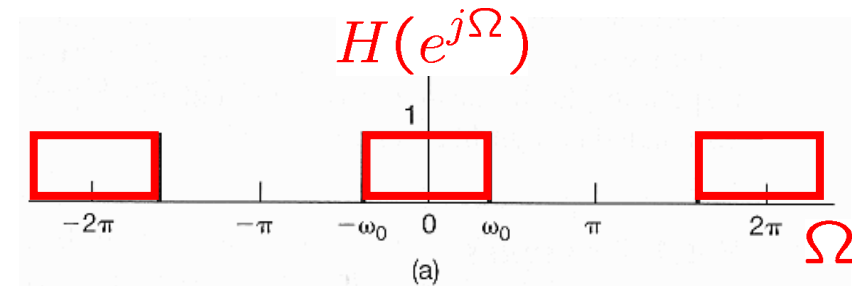
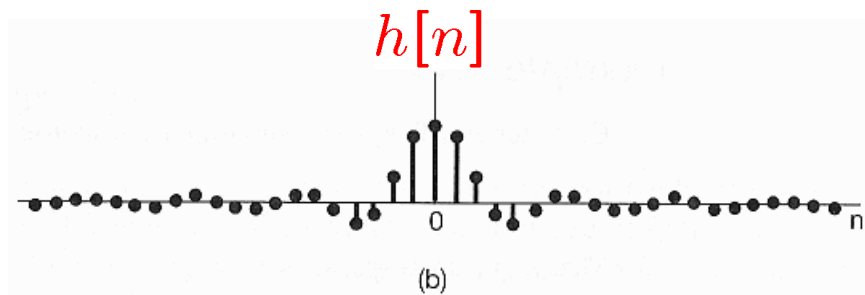


$$h(t) \xleftrightarrow{\text{C.T.F.T.}} H(j\omega)$$

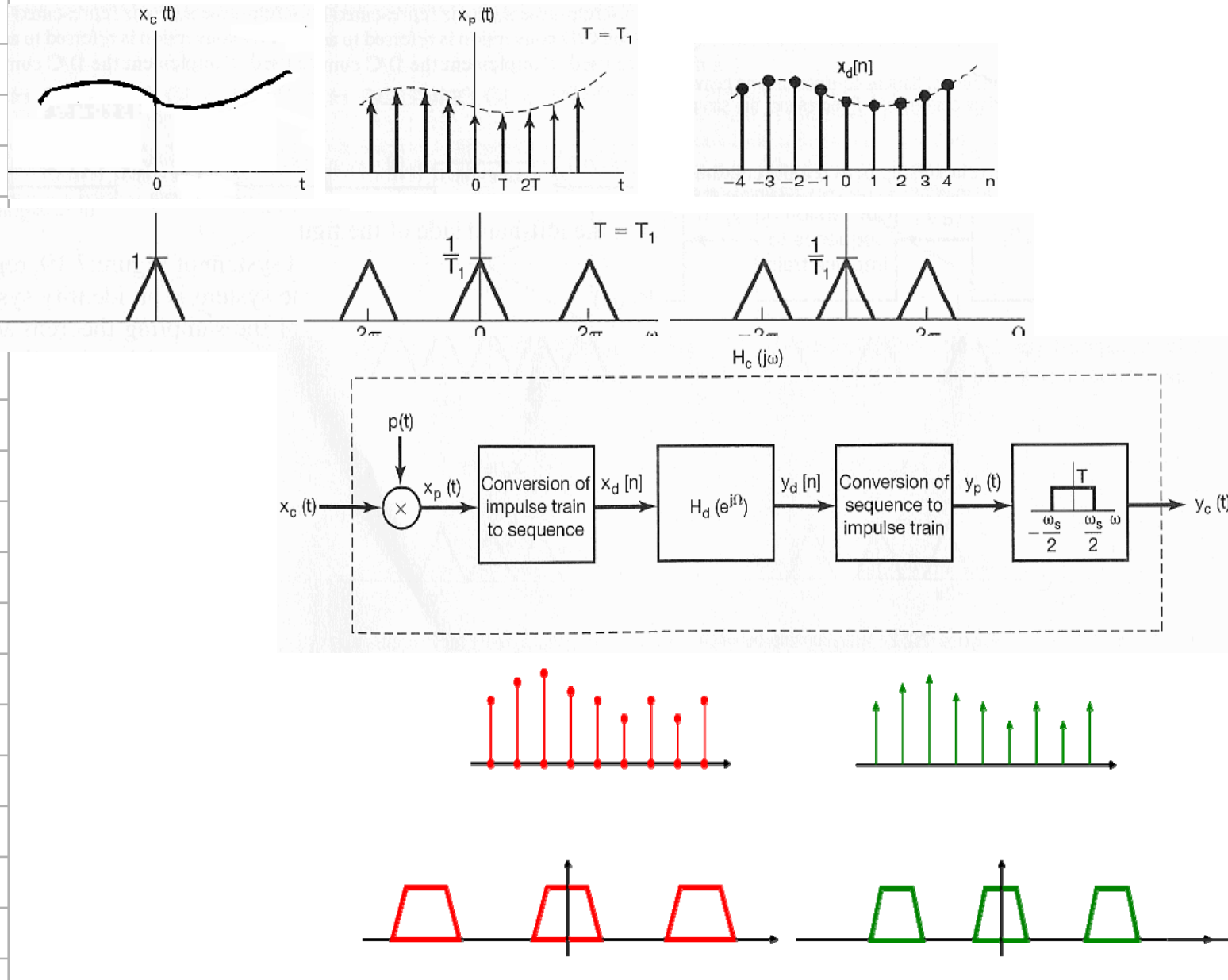
$$\omega_s = \frac{2\pi}{T}$$

$$\Omega = \omega T$$

$$h[n] \xleftrightarrow{\text{D.T.F.T.}} H(e^{j\Omega})$$



Discrete-Time Processing of CT Signals



Introduction

(Chap 1)

LTI & Convolution

(Chap 2)

Bounded/Convergent

Periodic

FS

(Chap 3)

– CT
– DT

Aperiodic

FT

– CT (Chap 4)
– DT (Chap 5)

Unbounded/Non-convergent

LT

– CT (Chap 9)

zT

– DT (Chap 10)

Time-Frequency (Chap 6)

CT-DT (Chap 7)

Communication (Chap 8)

Control (Chap 11)

Digital
Signal
Processing (dsp-8)